



# **AFRICA**

## **ENERGY FINANCE**

Business & Financial Models

### **BOOK 2**

# **Video Course Scripts**

Book 2. PAYGo Solar Finance

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# Contents

|                                                                                          |           |
|------------------------------------------------------------------------------------------|-----------|
| <b>About the course</b>                                                                  | <b>3</b>  |
| Running order                                                                            | 3         |
| Conventions                                                                              | 3         |
| Recordings                                                                               | 4         |
| <b>Module 0. Welcome: how the course, the book, the model and the tools fit together</b> | <b>5</b>  |
| <b>Module 1. The PAYGo business: retailer, utility and lender</b>                        | <b>9</b>  |
| <b>Module 2. Market, customers and affordability</b>                                     | <b>13</b> |
| <b>Module 3. The regulatory checklist</b>                                                | <b>17</b> |
| <b>Module 4. Product and price plan design</b>                                           | <b>21</b> |
| <b>Module 5. Distribution and customer acquisition</b>                                   | <b>25</b> |
| <b>Module 6. Repayment behaviour: cohorts, curves and defaults</b>                       | <b>29</b> |
| <b>Module 7. Portfolio KPIs and PAYGo PERFORM</b>                                        | <b>33</b> |
| <b>Module 8. Revenue recognition and credit losses</b>                                   | <b>37</b> |
| <b>Module 9. Unit economics</b>                                                          | <b>41</b> |
| <b>Module 10. Working capital, inventory and FX</b>                                      | <b>45</b> |
| <b>Module 11. Funding the book: equity, debt and receivables facilities</b>              | <b>49</b> |
| <b>Module 12. Securitisation and off balance sheet structures</b>                        | <b>53</b> |
| <b>Module 13. RBF, subsidies and affordability</b>                                       | <b>57</b> |
| <b>Module 14. Stress testing</b>                                                         | <b>61</b> |
| <b>Module 15. The investor view: due diligence and the investment memo</b>               | <b>65</b> |
| <b>Module 16. Case walk through: SolaraPay</b>                                           | <b>69</b> |
| <b>Module 17. Using MODEL 2, step by step</b>                                            | <b>73</b> |

# About the course

The video course follows the book chapter by chapter. Module 0 explains how the course, the book, the model, the case study, the templates and the decision tools fit together. Modules 1 to 16 follow Chapters 1 to 16, each built around the decision its chapter supports. Module 17 is a screen walk through of MODEL 2, the PAYGo Company Financial and Investment Model (version 0.8, development build), step by step, in the order of the user manual.

This document holds the full narration of every module, with the screen direction for each scene set apart in italics. The scripts are the reference text for the recordings and for subtitles, and they can be read on their own as a spoken summary of the book.

## Running order

| Module | Title                                                                   | Running time | Words narrated | Media                                            |
|--------|-------------------------------------------------------------------------|--------------|----------------|--------------------------------------------------|
| 0      | Welcome: how the course, the book, the model and the tools fit together | 10:43        | 1,442          | Narration, AEF_V2_Module_00.mp3                  |
| 1      | The PAYGo business: retailer, utility and lender                        | 10:41        | 1,399          | Narration, AEF_V2_Module_01.mp3                  |
| 2      | Market, customers and affordability                                     | 9:41         | 1,302          | Narration, AEF_V2_Module_02.mp3                  |
| 3      | The regulatory checklist                                                | 10:16        | 1,299          | Narration, AEF_V2_Module_03.mp3                  |
| 4      | Product and price plan design                                           | 9:53         | 1,299          | Narration, AEF_V2_Module_04.mp3                  |
| 5      | Distribution and customer acquisition                                   | 9:35         | 1,298          | Narration, AEF_V2_Module_05.mp3                  |
| 6      | Repayment behaviour: cohorts, curves and defaults                       | 9:50         | 1,300          | Narration, AEF_V2_Module_06.mp3                  |
| 7      | Portfolio KPIs and PAYGo PERFORM                                        | 11:28        | 1,481          | Narration, AEF_V2_Module_07.mp3                  |
| 8      | Revenue recognition and credit losses                                   | 10:31        | 1,363          | Narration, AEF_V2_Module_08.mp3                  |
| 9      | Unit economics                                                          | 9:50         | 1,283          | Narration, AEF_V2_Module_09.mp3                  |
| 10     | Working capital, inventory and FX                                       | 9:34         | 1,299          | Narration, AEF_V2_Module_10.mp3                  |
| 11     | Funding the book: equity, debt and receivables facilities               | 9:49         | 1,339          | Narration, AEF_V2_Module_11.mp3                  |
| 12     | Securitisation and off balance sheet structures                         | 9:53         | 1,306          | Narration, AEF_V2_Module_12.mp3                  |
| 13     | RBF, subsidies and affordability                                        | 10:01        | 1,347          | Narration, AEF_V2_Module_13.mp3                  |
| 14     | Stress testing                                                          | 9:43         | 1,294          | Narration, AEF_V2_Module_14.mp3                  |
| 15     | The investor view: due diligence and the investment memo                | 11:32        | 1,497          | Narration, AEF_V2_Module_15.mp3                  |
| 16     | Case walk through: SolaraPay                                            | 12:36        | 1,669          | Narration, AEF_V2_Module_16.mp3                  |
| 17     | Using MODEL 2, step by step                                             | 19:34        | 2,584          | Video, MP4, 1920 by 1080, subtitles and chapters |
| Total  |                                                                         | 3 h 15 min   | 25,801         |                                                  |

## Conventions

Every figure in the narration comes from the book, the model or the SolaraPay case, and the external figures keep the caveats the book attaches to them. SolaraPay Ltd and the Republic of Kivara are fictional, and their history is synthetic. The default model inputs describe a fictional market. Money is spoken in words with its currency, negative figures are spoken as a loss or as minus, and acronyms are said in full the first time they appear in a module.

Learning objectives are listed for the viewer and the instructor; they are not narrated. Each module closes with a recap and one practical exercise in the model, a template (T01 to T08) or a decision tool (D1 to D6).

## Recordings

The recordings of this edition use a British English narrator voice at a measured pace of about 130 words a minute, normalised to minus 16 LUFS for consistent loudness across modules. Each narration file carries its module title, author and track number. The Module 17 video is recorded from the model itself at 1920 by 1080 pixels, with chapter markers for each of the twelve steps and an English subtitle track, also supplied as a separate subtitle file.

# Module 0. Welcome: how the course, the book, the model and the tools fit together

Running time: 10:43. Book: Preface, How the book works with the model, Conventions, Status of this edition, and Chapter 1 section 1.7. Model sheets: Dashboard, Checks, Investment\_Readiness. Templates and tools: T01 to T08, D1 to D6.

## Learning objectives

1. State the central question of the course and the three businesses inside a PAYGo company.
2. Describe how the book, MODEL 2 (the PAYGo Company Financial and Investment Model), the SolaraPay case, the templates and the decision tools fit together.
3. Apply the course conventions and honesty rules when quoting any figure from the course.

## Scene 1. One question

On screen: *Title card. "Can PAYGo solar become profitable and financeable?"*

This course is built around one question. Can pay as you go solar, which we will call PAYGo from now on, become profitable and financeable?

The record of the sector so far suggests that the answer can be yes. But only for companies that understand what kind of business they are running. That understanding is the subject of everything that follows.

The course accompanies Book 2 of the Africa Energy Finance collection, "PAYGo Solar Finance: Business Models, Credit Risk and Financial Structuring for Solar Home Systems", by Emmanuel Boujeka Kanga. Each module follows one chapter of the book. Each one supports one decision.

## Scene 2. Three businesses in one

On screen: *Three columns. Retailer. Utility like service provider. Consumer lender.*

A PAYGo solar home system company, or SHS company, is three businesses at the same time.

It is a retailer, which books margin on the day of sale. It is a utility like service provider, which keeps the device working and the customer engaged for years. And it is a consumer lender. It waits for the cash, a few shillings or naira or kwacha a day, from households whose incomes rise and fall with harvests and remittances.

Many of the strategic and financial mistakes seen in this sector come from analysing one of those businesses and forgetting the other two. A company can look profitable on its income statement and still run out of cash. That is why this course reads PAYGo companies through customer cohorts and the receivables book, not through a simple sales line.

## Scene 3. Who the course is for

On screen: *Founders and CFOs. Credit officers. Impact investors. RBF programme managers. Graduate students.*

The course is written for the people who have to make decisions about these companies.

Founders and finance directors use it to build a plan that a lender will accept. Credit officers at banks and development finance institutions use it to size a facility. Investment managers at impact funds use it to prepare an investment committee memo. Programme managers use it to design results based financing, which we will call RBF.

It is also meant for graduate students who want to understand how distributed energy access is financed, beyond the headline numbers.

#### Scene 4. The map of the course

On screen: *Course map. Module 1 foundations. Modules 2 to 5 commercial. Modules 6 to 9 credit and profitability. Modules 10 to 13 cash and capital. Modules 14 to 16 risk, diligence and case. Module 17 model walk through.*

The book has sixteen chapters, and the course has a module for each.

Module 1 sets out the business model, and explains why cohorts and the receivables book are the unit of analysis.

Modules 2 to 5 cover the commercial foundations. These are customers and affordability, regulation, product and price plan design, and distribution.

Modules 6 to 9 deal with credit and profitability. You will study repayment behaviour and the key performance indicators, or KPIs, of a portfolio. You will then study revenue recognition, credit losses and unit economics.

Modules 10 to 13 turn to cash and capital. They cover working capital and currency, the funding stack, securitisation and subsidies.

Modules 14 and 15 bring the analysis together for the people who carry the risk, through stress testing and the investor's diligence and memo. Module 16 walks through a complete case. A final module then walks you through the workbook itself, sheet by sheet.

#### Scene 5. The companion workbook

On screen: *MODEL 2, PAYGo Company Financial and Investment Model, v0.8 (development build). 47 sheets. About 100,900 formulas. 38 integrity tests. Calculation chain from Inputs to Investment readiness.*

The book has a companion workbook, MODEL 2, the PAYGo Company Financial and Investment Model. This course uses version 0.8, a development build.

It is a monthly, five year, integrated three statement model. It has 47 sheets and about 100,900 formulas, with no macros and no circular references. Its Checks sheet runs 38 integrity tests. Its outputs have been checked three ways: against a secondary calculation written separately, which agrees to rounding precision, by an independent formula engine, and by a full recalculation in LibreOffice. A test in Microsoft Excel is still pending.

The calculation chain runs from Inputs and Products, through Sales, Customers, PAYGo receivables, the Credit engine, the Vintage engine, and Collections and recoveries. It continues to the RBF engine, Working capital, Financing and the Financial statements. It ends with Covenants, Valuation, Investor returns, the Market benchmark, Calibration and Investment readiness.

The model carries five product tiers. They run from a pico solar kit on a twelve month plan to a large solar inverter with lithium storage on a forty eight month plan.

The chapters refer to the workbook by sheet name, so that each concept can be tested on numbers. You can follow the argument without opening it. If you do work through the sheets, you will finish the course able to build, challenge and defend a PAYGo investment case.

## Scene 6. The SolaraPay case

On screen: *SolaraPay Ltd, Republic of Kivara. Fictional company. Synthetic 24 month history. Ask: 4 million dollars equity, 4.5 billion shilling facility.*

A worked case runs through every module. SolaraPay Ltd is a fictional company in the fictional Republic of Kivara, with a synthetic twenty four month history.

SolaraPay is asking an impact fund for four million dollars of equity at a pre money valuation of eight million dollars. It is also asking a local bank for a receivables facility of four point five billion shillings.

Its management plan looks attractive. Once the model is calibrated to its own history, the case becomes conditional. The way it moves from one to the other is the method this course teaches.

The workbook never rates a company. Its Investment\_Readiness sheet runs 23 gates, counts each one only on evidence, and applies a fixed rule. It reads STOP when a test fails or critical evidence is missing, CONDITIONAL GO when every critical gate is met, and GO only when all 23 are met. Even a GO says only that the evidence file is complete. It is not a recommendation. In its default state the workbook passes 2 of the 23 gates and reads STOP on a failed test: the annual debt service coverage ratio sits below its minimum in Years 1 to 4, so the covenant gate fails. SolaraPay, with its history loaded, passes 4 and reads STOP for the same reason. The analyst in the case still recommends a Conditional Go, and Module 16 explains why the two do not contradict each other.

## Scene 7. Templates and decision tools

On screen: *Templates T01 to T08. Decision tools D1 to D6. Book, tools, model: three levels of depth.*

Two kits sit beside the book and the model.

The templates pack holds eight working documents, numbered T01 to T08. They are the documents a deal team exchanges during a transaction. They include the business model canvas, the loan tape and data request, the due diligence checklist, the lender KPI report, the borrowing base certificate, the customer credit policy, the agent economics sheet and the investment committee memo.

The decision tools are six compact calculators, numbered D1 to D6. Each one answers a single question, such as whether a price plan is affordable, or what an entry price implies in dollars. They sit between the book and the full model. The book explains the reasoning. The tools give a fast answer you can check by hand. The model puts everything together over five years.

## Scene 8. Conventions and honesty

On screen: *Fictional inputs. Sourced external figures with caveats. Conflicts reported, not resolved. Pre-publication edition.*

A few conventions apply throughout.

All default inputs in the workbook, and all SolaraPay figures, are illustrative. They were chosen to make the mechanics visible and to resemble the orders of magnitude found in the sector. They do not describe any real company. In the generic model, amounts are in local currency units. In the case, they are in shillings.

Where the course cites figures about real companies or about the sector, it says where they come from and how far they have been verified. Several published figures conflict with each other. When they do, the course reports the conflict rather than choosing the more convenient number.

The course follows the first edition of Book 2, issued as version 0.2 for pre-publication review. It will be revised after the workbook completes its test in Microsoft Excel, after the primary documents behind the claims marked as pending have been read and cited by page, and after practitioner review. Claims marked as pending or conflicting in Annex G of the book must not be quoted without checking the original source.

Nothing in the course is investment, legal, tax or accounting advice. Take professional advice in the jurisdiction concerned before acting on any of it.

## Scene 9. Recap and exercise

On screen: *Exercise. Open the workbook. Checks sheet. Dashboard. Investment\_Readiness: 2 of 23, STOP on a failed test.*

To recap. The course asks whether PAYGo solar can become profitable and financeable. It treats each company as a retailer, a service provider and a lender at once, and it reads that company through cohorts and the receivables book. The book gives the reasoning, the workbook tests it on numbers, and the templates and decision tools carry it into a real transaction.

Your exercise before Module 1 is practical. Open MODEL 2 in its default state. Go first to the Checks sheet and confirm that it reads OK. Then open the Dashboard and note the peak equity and the investor internal rate of return, or IRR, for the Base case. Finally, open Investment\_Readiness, confirm that it shows 2 of 23 gates passed and a decision of STOP on a failed test, and write down the name of the failed test and of two other gates that fail. You will return to that list at the end of the course.

# Module 1. The PAYGo business: retailer, utility and lender

Running time: 10:41. Book: Chapter 1. Model sheets: Dashboard, KPIs, Unit\_Economics, Credit\_Portfolio, Vintage\_Dashboard, Checks, Investment\_Readiness. Templates and tools: T08 PAYGo business model canvas, D2 Unit economics calculator.

## Learning objectives

1. Separate a PAYGo company into its retail, service and lending businesses and name the numbers that belong to each.
2. Reconcile the day one margin of a single sale with its cash flow and its lifetime accounting result.
3. Explain why cohorts and the receivables book, not the sales line, are the unit of analysis, and recognise the three common failure modes.

## Scene 1. The decision

On screen: *Title. "What business is a PAYGo company really in?"*

This module supports one decision, which comes before any other. What business is a PAYGo, or pay as you go, solar company really in? And therefore, which numbers should an investor, a lender or a board read first?

Funding follows analysis. A company analysed as a retailer will be funded as a retailer. And a retailer's funding will not carry a two year consumer credit book.

## Scene 2. Three businesses in one legal entity

On screen: *Three panels. Retailer: gross margin. Service provider: cost to serve. Lender: receivables and credit losses.*

The customer pays a deposit, takes the system home and pays a daily rate through mobile money for a fixed tenor. At the end, the device unlocks for good. Seen from outside, that is one business. Inside, it is three.

The first is a retailer. It buys hardware, runs agents and earns a gross margin.

The second is a utility like service provider. For two to four years it serves each active account, at a cost largely independent of the original sale.

The third is a consumer finance company. Every unit sold creates a receivable, in effect an unsecured loan to a household with irregular income. It must fund that book.

Read as a retailer, the business looks highly profitable. Read as a lender, it looks like a high risk portfolio. Each reading is partly right. The investment case depends on how the three interact.

## Scene 3. The lockout is not security

On screen: *Lockout. Daily repayment decision. Weakens late in tenor. No recovery value. SolaraPay loss given default about 99%.*

The lockout does not secure the receivable the way a charge over an asset secures a loan. It gives the customer a reason to keep paying, every few days. Collection behaviour becomes visible quickly. The deterrent weakens as the tenor ends and the battery ages. And the lockout creates no recovery value. The company recovers money only if it can retrieve, refurbish and resell the device for more than that costs. In the SolaraPay case, the observed proxy for loss given default is about 99 per cent.

At unlock the customer becomes an owner. Many companies cannot evidence how many reach ownership, and the workbook will not invent that number.

#### Scene 4. One sale, two stories

*On screen: Worked example, Tier 2 SHS with TV. Cash price 40,000. Deposit 4,000. Daily rate 72. Tenor 24 months. Landed cost 22,000. Acquisition cost 4,000.*

Take one illustrative sale of a solar home system with a television in the fictional model market.

The cash price is forty thousand local currency units. The deposit is four thousand. The daily rate is seventy two, which gives a monthly instalment of seventy two times 365 divided by twelve, or two thousand one hundred and ninety. Over twenty four months that is fifty two thousand five hundred and sixty of scheduled instalments. With the deposit, the contract is worth fifty six thousand five hundred and sixty. The PAYGo premium, the financing charge, is sixteen thousand five hundred and sixty. The hardware costs twenty two thousand landed, and acquiring the customer costs four thousand.

On the income statement, day one looks good. Hardware revenue of forty thousand, less cost of sales of twenty two thousand, gives a gross margin of eighteen thousand, or 45 per cent. After acquisition cost, the contribution is fourteen thousand. Financing income of six hundred and ninety a month follows over the tenor.

In cash, day one looks very different. The company has paid twenty two thousand for hardware and four thousand for acquisition, and received four thousand of deposit. It is twenty two thousand local currency units out of pocket.

#### Scene 5. The repayment curve

*On screen: Chart. Cumulative cash per unit by month. Minus 22,000 at month 0, crossing zero at month 14, 11,832 at month 24.*

The workbook drives repayment with two parameters. The monthly default hazard, 2.6 per cent for Tier 2, is the share of paying accounts that stop for good each month. The collection rate on paying accounts, here 88 per cent, reflects the days even good customers miss.

Over the full tenor, expected collections come to thirty three thousand eight hundred and thirty two. That is 64.4 per cent of the scheduled instalments.

The unit recovers its cash outlay in month fourteen. Over the tenor it returns eleven thousand eight hundred and thirty two in cash. And that is before mobile money fees, servicing, warranty and the cost of funding a deficit for more than a year.

#### Scene 6. Where accounting and cash meet

*On screen: Day one result with provision: a loss of 4,728. Lifetime result: 11,832. Without provision: 14,000 profit on day one.*

Expected missed instalments over the contract are fifty two thousand five hundred and sixty less thirty three thousand eight hundred and thirty two. That is eighteen thousand seven hundred and twenty eight.

The workbook charges that amount as a loss allowance at origination. So the day one result is eighteen thousand, less four thousand, less eighteen thousand seven hundred and twenty eight. That is a loss of four thousand seven hundred and twenty eight. Financing income of sixteen thousand five hundred and sixty then brings the lifetime result to eleven thousand eight hundred and thirty two. Accounting and cash reconcile.

The workbook's treatment is a stated simplification of the international standards, IFRS 15 and IFRS 9, chosen because it makes each cohort's accounting and cash agree exactly. This is an analytical modelling treatment and does not constitute a determination of the applicable accounting treatment under IFRS.

A company that recognises losses only as accounts fall behind reports fourteen thousand of contribution on day one. When sales grow fast, margin on new sales dominates, and losses belong to older, smaller cohorts. Reported profit looks healthy. Cash is still minus twenty two thousand per unit. A company selling three thousand such units a month needs sixty six million local currency units of new funding every month.

That is the gap. Everything that matters in the business sits in it.

## Scene 7. Cohorts and the receivables book

On screen: *Cohort curves against plan at the same age. Receivables book: largest asset, collateral, main use of equity. Vintage\_Dashboard.*

Any month's results blend cohorts of different ages. A falling portfolio collection rate can reflect weaker credit, faster growth, or both. Without cohort data you cannot tell which.

So the course treats the cohort, or vintage, as the primary object of analysis. A cohort is the contracts originated in the same month, by tier. Its repayment curve, read against plan at the same age, is the most direct evidence that underwriting works. The receivables book is the company's largest asset and the collateral for its debt.

PAYGo PERFORM is the industry's reporting standard. Its current version, the technical guide published by GOGLA in June 2026, defines five KPIs, or key performance indicators, on repayment and ownership. They are computed by the company on its own contract data. The workbook keeps the older ratios, such as the collection rate, as operational and lender metrics, each with its definition printed, and reports PERFORM 2026 figures only when the company supplies them. The sector evidence is sobering, although it still has to be read in the original. The ESMAP Off-Grid Solar Market Trends Report 2024 is reported to give an average PAYGo collection rate of about 62 per cent for 2021 to 2023. That figure is pending the primary document, and it is a collection rate, not a PERFORM repayment rate.

## Scene 8. How PAYGo companies fail

On screen: *Three failure modes. Growth outruns funding. Credit drift. Foreign exchange.*

The failure modes are few, and they compound.

The first is growth outrunning funding. In the workbook's default case, the fictional company needs peak equity of ten million dollars. In the Severe case that rises to forty six point six million dollars.

The second is credit drift. In the SolaraPay case, the portfolio collection rate fell from 77.4 per cent in the first history year to 69.3 per cent in the second.

The third is foreign exchange, or FX, because companies borrow in dollars and collect in local currency. In the default sensitivities, depreciation of 20 per cent a year turns a Base investor internal rate of return, or IRR, of 46.4 per cent into minus 16.0 per cent. Freezing prices on new contracts cuts it to 7.9 per cent.

None of these shows first on the income statement. The sector has seen distress. BBOX LTD, a UK company, is recorded as having entered administration on 19 May 2025. The Gazette notice and the register entry have not yet been read, so the date is pending the primary document. The course does not speculate on the causes.

## Scene 9. Recap and exercise

On screen: *Exercise. T08 canvas. Three businesses. Day one margin against cash. Unit\_Economics and D2.*

To recap. A PAYGo company is a retailer, a service provider and a lender in one entity. Its day one margin and its cash diverge for more than a year per unit, and the gap grows with growth. Read the cohort curves, the funding plan and the FX bridge before the income statement.

Your exercise uses the T08 PAYGo business model canvas. Fill it for a company you know, or for SolaraPay, with gross margin, cost to serve per active account and credit losses in separate blocks. Then open Unit\_Economics in the workbook, or the D2 unit economics calculator, and find the month in which each tier recovers its cash outlay. Write down which tier takes longest, and how much funding each unit needs until then.

## Module 2. Market, customers and affordability

Running time: 9:41. Book: Chapter 2. Model sheets: Consumer\_Risk, Products, Scenarios, Checks, Investment\_Readiness. Templates and tools: D1 Price plan and APR calculator, T03 Customer credit policy.

### Learning objectives

1. Match each product tier to its customer segment and explain how seasonality changes the affordability test.
2. Compute the payment burden, the deposit share of income and the implied APR of a price plan, under both the nominal and effective conventions.
3. Judge the evidence behind affordability data and link a high burden to the default hazard assumptions in the model.

### Scene 1. Who can pay, and how much

On screen: Title. "Who can pay, and how much?"

This module supports one decision: who can pay, and how much.

Every PAYGo, or pay as you go, price plan bets that a household can meet an instalment every month, for the whole tenor, out of a small and irregular income.

Counting households without grid access answers a different question. The investor must know how many can carry the instalment, and the lender how that changes in a bad harvest.

### Scene 2. Segments by tier

On screen: Table of five tiers. Pico kit, 12 months. SHS with TV, 24 months. SHS with fridge, 30 months. Inverter 1.2 kWp, 36 months. Inverter 3 kWp, 48 months.

The workbook carries five product tiers, and each implies a different customer.

Tier 1 is a pico solar kit on a twelve month plan, often for rural customers new to formal credit. The instalment competes with kerosene and torch batteries.

Tier 2 is a solar home system with a television, over twenty four months. When money is short, television is easier to give up than light.

Tier 3 adds a fridge, over thirty months, and can turn the product into an income source for a kiosk.

Tiers 4 and 5 are solar inverters with lithium storage, over thirty six and forty eight months. They serve urban households and small businesses on a weak grid, with higher incomes. But the amounts financed are an order of magnitude larger, and for most companies their repayment behaviour is unknown.

The default monthly hazard falls as tiers rise, from 3.5 per cent for Tier 1 to 2.6, 1.9, 1.3 and 1.0 per cent for Tiers 2 to 5. These are model defaults for a fictional market, not sector evidence. Replace them with cohort data where you have it.

### Scene 3. Seasonality

On screen: *Income through the year. Harvest peak. Lean months. School fees.*

Most rural customers earn from farming, casual labour, trading or remittances. Cash is plentiful after harvest and scarce in the months before it, when food runs low and school fees fall due.

This has three consequences. An affordability ratio on average income overstates capacity in the months that matter. A month's collection rate must be compared with the same month in earlier years before it is read as a trend. And cohorts sold in different seasons behave differently. A customer who buys just after harvest pays the deposit easily, then meets the lean season a few months in.

Version 0.8 of the workbook works with monthly averages and does not model seasonality. You must apply it yourself, in the affordability test and in reading monthly figures.

### Scene 4. Payment burden

On screen: *Payment burden equals monthly instalment divided by monthly household income. Threshold 10%. Consumer\_Risk sheet.*

The core measure is the payment burden. It is the monthly instalment divided by monthly household income. The Consumer\_Risk sheet computes it for each tier and raises a flag when it exceeds a maximum. The default maximum is 10 per cent.

That threshold is a model policy threshold, not a law, a regulatory standard or an empirical boundary between good and bad credit. A company may justify a different line with repayment evidence. What it should not do is set the threshold after the fact to clear the products it wants to sell.

Take the illustrative Tier 2 plan from Module 1. The instalment is two thousand one hundred and ninety local currency units. Assume average monthly income of eighteen thousand, made of six lean months at twelve thousand and six good months at twenty four thousand.

On average income the burden is 12.2 per cent, so the plan fails. In a good month it is 9.1 per cent and passes. In a lean month it is 18.3 per cent.

To pass on average income, the household must earn at least twenty one thousand nine hundred a month. Or the instalment must fall to about one thousand eight hundred, which means a daily rate of about fifty nine.

SolaraPay shows the same problem in its own plan. On its illustrative incomes, the burden is 13.0 per cent for Tier 2, 13.2 per cent for Tier 3 and 11.3 per cent for Tier 4. All three are above the threshold.

### Scene 5. The deposit as a screen

On screen: *Deposit: lowers exposure, screens customers. 22.2% of average income. 33.3% of a lean month.*

The deposit reduces the amount financed. More importantly, it screens customers. A household that can assemble a few weeks' income has shown it can save.

A higher deposit lowers conversion, and the customers who remain should repay better. Only a company that has varied its deposit and tracked the cohorts knows where the balance lies.

In our example, the deposit of four thousand is 22.2 per cent of average monthly income, and 33.3 per cent of a lean month. Watch for deposits cut sharply in sales campaigns. Weaker cohorts tend to follow, and portfolio figures hide the effect for several months.

## Scene 6. The implied APR

On screen: *Tier 2 example. Monthly rate 3.28%. Nominal APR 39.3%. Effective rate 47.3%. Flat rate 23.0%. D1 calculator.*

Customers see a deposit, a daily rate and a number of days, not a loan. The implied annual percentage rate, or APR, is the figure a regulator or a journalist will compute. Management should know it first.

The monthly rate is the one that discounts the instalments back to the amount financed. For the Tier 2 plan, thirty six thousand financed, repaid at two thousand one hundred and ninety a month over twenty four months, it is about 3.28 per cent.

Twelve times that monthly rate gives a nominal APR of 39.3 per cent. Compounding it over a year gives an effective rate of 47.3 per cent. Any disclosure must say which convention it uses, and local counsel must confirm which one the rules require.

The flat rate is the premium divided by the amount financed and by the years. Here it is 23.0 per cent. It understates the cost by about half, because the balance declines as the customer pays.

SolaraPay's implied APRs run from 106 per cent on Tier 1 down to 50, 39, 32 and 25 per cent. The highest rate sits on the smallest product, because fixed costs are spread over a small amount financed. It is also the most exposed to a rate cap.

## Scene 7. The Multi Tier Framework and the data

On screen: *Multi Tier Framework attributes. Capacity, duration, reliability, quality, affordability, legality, health and safety. Data: surveys, agents, mobile money, own repayment.*

The Multi Tier Framework from ESMAP, set out in Beyond Connections in 2015, defines Tiers 0 to 5 of household electricity access. It uses capacity, duration, reliability, quality, affordability, legality, and health and safety. The workbook uses the tier labels on capacity alone. Treating a capacity label as a full access assessment in impact reporting is a common error.

The illustrative incomes in the workbook are placeholders. A readiness flag on the Checks sheet stays raised until you replace them and record their status.

Surveys, agent data and mobile money histories can fill them, each with its own bias. In the end, the company's own repayment data are the most reliable test. A customer who pays for twelve months has shown the instalment was affordable over a full seasonal cycle.

## Scene 8. From affordability to default risk

On screen: *SolaraPay Tier 2. Burden 13.0%. Hazard 2.60% to 3.38%. Month 12 repayment 69.4% against 74.5% plan.*

Affordability is a credit risk before it is a social one. The workbook does not link burden to hazard by formula. You make that link through the hazard inputs on Products, and test it on Scenarios.

SolaraPay shows why. Its Tier 2 burden is 13.0 per cent. Calibrated to its history, the Tier 2 monthly hazard rises from 2.60 to 3.38 per cent. Cumulative repayment at month twelve is 69.4 per cent, against a plan of 74.5. The data do not prove that the burden caused the shortfall, but they are consistent with it. An investment committee would not accept a plan that assumed the problem away.

### Scene 9. Recap and exercise

On screen: *Exercise. D1 calculator, Plan A. Lean month burden. Daily rate for 10%. Nominal and effective APR.*

To recap. Affordability is measured against the instalment, by tier, and on lean month income as well as the average. The 10 per cent threshold is a policy line that must be evidenced. The deposit screens as well as reduces exposure, and the implied APR must be stated on a declared convention.

Your exercise uses the D1 price plan and APR calculator. Open Plan A, the illustrative Tier 2 plan. Change the deposit to eight thousand and the tenor to thirty months. Record the new instalment, the burden on average and on lean month income, and both APR figures. Then use the solver block to find the daily rate that meets the 10 per cent threshold, and note what that rate does to the PAYGo premium.

## Module 3. The regulatory checklist

Running time: 10:16. Book: Chapter 3. Model sheets: Products, Inputs, Scenarios, Credit\_Assumptions, Investment\_Readiness. Templates and tools: T02 Investor due diligence checklist, D1 Price plan and APR calculator.

### Learning objectives

1. Name the rule categories that apply to a PAYGo company and the question to ask in each.
2. Translate each regulatory risk into the line of the model that moves, and test it through inputs and scenarios.
3. List the evidence an investor should expect, and decide which legal positions must be conditions precedent.

### Scene 1. What could stop the company

On screen: Title. *"What could stop the company operating, or stop it scaling?"*

This module supports one decision: what could stop the company operating, or stop it scaling.

A PAYGo, or pay as you go, company sits under several regimes at once. Consumer credit, payments, data protection, tax, product standards, environmental rules and exchange control. Each can change the economics. Some can halt the business.

The module is a checklist of rule categories. For each one you will see the question to ask, why it matters financially and what evidence to expect. It names no country's law and no regulator's rule. Rules differ by jurisdiction and change often. In every category, the position must be confirmed with local counsel before an investment committee relies on it.

### Scene 2. How to use the checklist

On screen: *Licence: cost and timing. APR cap: price plan. Lockout ban: default hazard. Exchange control: debt service.*

Regulatory diligence is often done late and narrowly. A better approach treats each rule as a driver of the financial model.

A licence requirement is a cost and a timing risk. A cap on the annual percentage rate, or APR, constrains the price plan. A ban on remote lockout changes the default hazard. An exchange control constrains dividends and debt service.

So the question is always the same. If the rule applies, or changes, which line of the model moves, and by how much? Where the answer is material, the committee should see the scenario run, not only the legal memo.

Three general points apply. The relevant law is that of each operating country. The perimeter around PAYGo is often unclear, because the product sits between a hire purchase sale, a consumer loan and a utility service. And unclear rules can become clear quickly, rarely in the company's favour.

### Scene 3. Credit licences and pricing

On screen: *Is the contract credit? Who needs a licence? Disclosure. APR caps. Tier 2 example: flat 23.0%, effective 47.3%.*

The first question is whether the PAYGo contract counts as credit under local law. If so, the company, or the entity holding the receivables, may need a lending, credit or hire purchase licence.

Without a required licence, contracts may be unenforceable, collections may be challenged and lenders' security may be impaired. With one come capital requirements and reporting. In a group, the same question applies to a financing subsidiary or a special purpose vehicle, an SPV.

Next comes pricing disclosure and APR caps. A company that has not computed its own APR will learn it from someone else. The illustrative Tier 2 plan has a flat rate of 23.0 per cent a year, but an effective annual rate of 47.3 per cent. SolaraPay's implied APRs run from 25 per cent on Tier 5 to 106 per cent on Tier 1.

A cap hits small, short products hardest. If Tier 1 breaches it, the company can raise the cash price, which shifts margin from financing income to hardware revenue without changing what the customer pays. Or it can accept a lower return, or withdraw the product. Model each option.

#### Scene 4. Collection conduct and lockout

On screen: *Lockout rules. Notice periods. Repossession. Agent incentives. Complaints log.*

The lockout is the company's main collection tool. So ask whether there are rules on remote lockout, notice periods, repossession and conduct towards customers.

A rule that restricts disablement, or requires a grace period, changes the default hazard directly. A longer notice period before lockout will, other things equal, raise the share of customers who fall behind. A ban on repossession removes the recovery line altogether. In many portfolios that line is already small. SolaraPay's observed proxy for loss given default is about 99 per cent.

Conduct also carries licence and reputational risk. Aggressive collection by agents paid on recoveries can lead to sanctions, and to exclusion from results based financing, or RBF, and from development finance with consumer protection requirements.

Ask for the collections policy, the lockout procedures and the complaints log.

#### Scene 5. Data and payments

On screen: *Consent. Cross border transfer. Credit bureau. Mobile money fees. Settlement times.*

Data protection asks what consent the company needs to use customer data, including mobile money histories and device usage, and whether data can cross borders. A restriction on using mobile money histories removes one of the strongest affordability sources. A limit on cross border transfer can affect a group that runs its credit platform from another country.

Credit bureau reporting cuts both ways. It can improve discipline and let good payers build a history that supports upgrades. It also brings compliance cost.

Almost all collections arrive through mobile money. In the workbook, operator fees are an input, as a percentage of cash collected. Settlement terms decide how quickly cash reaches the bank account, and so how much sits outside the borrowing base. Ask for every operator agreement, with fees, settlement times, exclusivity and termination clauses.

## Scene 6. Tax, standards and e-waste

*On screen: VAT and duty on solar. Withholding. Transfer pricing. Tax 30%, losses carried forward. Product quality. Battery take back.*

Many PAYGo cost structures rely on favourable VAT and duty treatment of solar products. Losing it raises the landed cost of every new unit. The workbook carries freight, duty and clearing as a percentage of the free on board cost, so you can run the change directly.

Withholding tax on interest raises the cost of offshore debt where the lender needs a gross up. On dividends, it cuts what the equity investor receives. Groups must also document transfer prices for hardware and services.

The workbook applies corporate tax at 30 per cent, with losses carried forward without limit. Confirm that locally. A cap on carry forward brings tax forward in a business that loses money in its early years.

Quality matters everywhere in the model. Customers stop paying for devices that do not work. Products that miss a standard may be blocked at import or excluded from RBF. Programmes that pay per verified connection, such as SEforALL's Universal Energy Facility, set their own eligibility conditions.

Batteries are hazardous at end of life. A take back obligation creates a cost that may fall years after the sale. For Tiers 4 and 5, with large lithium batteries, that cost is material.

## Scene 7. Exchange control and securitisation

*On screen: Convert and remit: suppliers, debt, dividends, exit. 20% depreciation: IRR 46.4% to minus 16.0%. True sale opinion.*

For an offshore investor, exchange control is often among the most important questions. Can the company convert local currency and remit it, for hardware, debt service, dividends and exit proceeds? If conversion is rationed or delayed, a company solvent in local currency may still fail to pay suppliers or lenders. Registration of foreign investment at entry is often a condition for repatriation later, so it cannot be left to exit.

The price risk is already severe. At default inputs, depreciation of 20 per cent a year turns a Base investor internal rate of return, or IRR, of 46.4 per cent into minus 16.0 per cent. Controls add a liquidity risk on top.

Securitisation lets a company fund its book at a scale a balance sheet cannot reach. Sun King is reported to have completed local currency securitisations in Kenya of about one hundred and thirty million dollars in May 2023, and about one hundred and fifty six million dollars in July 2025. That is company and secondary reporting. d.light is reported to have about seven hundred and eighteen million dollars of purchasing capacity across five facilities since 2020. That is secondary reporting, and it is purchasing capacity, not debt raised.

The legal questions are specific. Can receivables be assigned without customer consent? Does the sale survive the originator's insolvency? Can the originator keep servicing, including the lockout? In the workbook's default case, securitisation leaves the investor IRR almost unchanged, at 46.5 per cent against 46.4. Its value lies in scale, tenor and cost of funding.

## Scene 8. Recap and exercise

*On screen: Exercise. T02 checklist. Regulatory map by category. Lockout restriction run on Products and Scenarios.*

To recap. Treat every rule category as a driver of the model, and ask which line moves if the rule applies or changes. Expect evidence, not assertions, and have every position confirmed by local counsel. Where securitisation is in the funding plan, make a true sale opinion a condition precedent, not a post closing item.

Your exercise has two parts. First, using the legal and regulatory workstream of the T02 due diligence checklist, build a one page map for SolaraPay listing each category, its financial impact and the evidence still missing. Second, assume a new rule that delays lockout. Raise the Tier 1 and Tier 2 default hazards on the Products sheet by 30 per cent, and record the change in investor IRR and covenant breach months.

## Module 4. Product and price plan design

Running time: 9:53. Book: Chapter 4. Model sheets: Products, Consumer\_Risk, Unit\_Economics, Sensitivity, Scenarios, Vintage\_Dashboard. Templates and tools: D1 Price plan and APR calculator, D2 Unit economics calculator, D3 Repayment curve and cohort calibration.

### Learning objectives

1. Derive the instalment, contract value, premium and amount financed from the four price plan inputs, and explain how the cash price moves revenue and APR.
2. Compute the expected collected share of a plan from the hazard, the collection rate and the tenor, and use it to compare tenors.
3. Assess product mix, upgrades and warranty, and decide how to treat untested tiers.

### Scene 1. The design decision

On screen: Title: *"Which price plan maximises value without destroying repayment?"*

This module supports one decision: which price plan maximises value without destroying repayment.

A price plan has five components. The cash price, the deposit, the daily rate, the tenor, and the PAYGo, or pay as you go, premium that results. Each moves revenue, exposure, affordability and default risk at once.

The plan that maximises contract value is rarely the one that maximises cash collected.

### Scene 2. The components

On screen: *Products sheet. Four inputs: cash price, deposit, daily rate, tenor. Derived: instalment, contract value, premium, amount financed, implied APR.*

The Products sheet takes four inputs for each tier: cash price, deposit, daily rate and tenor. The instalment, contract value and premium are derived from them.

The cash price matters more than it looks. Hardware revenue equals the cash price and is booked at sale. The premium is spread over the tenor as financing income. So a high cash price with a low premium brings revenue forward, without changing what the customer pays. It also lowers the implied annual percentage rate, or APR. Check that the cash price is one at which the company really sells for cash.

A common design error is to set the daily rate from a target contract value, and only then ask whether the customer can pay.

### Scene 3. Tenor and deposit

On screen: *Longer tenor: lower instalment, more exposure, more months at risk. Higher deposit: screening, lower conversion. Deposit doubled: amount financed down 11.1%.*

Two trade offs dominate design.

A longer tenor lowers the instalment, and so the payment burden. It also raises exposure and funding per unit. With a monthly hazard, every extra month is another month in which the account can stop.

A higher deposit screens customers and lowers exposure. In the illustrative Tier 2 plan, doubling the deposit from four thousand to eight thousand local currency units cuts the amount financed by 11.1 per cent. It also lifts the deposit from 22.2 to 44.4 per cent of average monthly income, and from 33.3 to 66.7 per cent of a lean month. Conversion will fall, and only tested cohorts show whether better repayment compensates.

A higher premium lifts contract value, but also the burden and the APR. If the hazard rises with it, part of the extra premium is never collected.

#### Scene 4. How collection falls with tenor

On screen: *Survival 0.974 to the power of age. Month 6: 0.854. Month 12: 0.729. Month 18: 0.622. Month 24: 0.531. Expected collected share: 18 months 69.2%, 24 months 64.4%.*

With a monthly hazard of 2.6 per cent, the share of Tier 2 accounts still paying is 0.974 raised to the power of the account's age in months.

At month six, about 85 per cent are still paying. At month twelve, about 73 per cent. At month twenty four, about 53 per cent.

The expected share of scheduled instalments collected is the collection rate on paying accounts, times the average survival over the tenor. With a collection rate of 88 per cent, an eighteen month plan collects an expected 69.2 per cent of its instalments. A twenty four month plan collects 64.4 per cent.

So moving from eighteen to twenty four months costs 4.8 percentage points of expected collection. The last six months of the longer plan carry 25 per cent of the scheduled instalments, but only 19.4 per cent of the expected collections. The company is lending into months in which many customers have already stopped paying.

#### Scene 5. No easy way out through tenor

On screen: *Same expected cash. 18 months: instalment 2,718, burden 15.1%. 30 months: instalment 1,879, burden 10.4%.*

The same arithmetic shows what a shorter tenor costs in affordability. The Tier 2 plan expects to collect thirty three thousand eight hundred and thirty two over twenty four months. To collect the same over eighteen months, the instalment must rise to about two thousand seven hundred and eighteen. On average income of eighteen thousand, the burden rises from 12.2 to 15.1 per cent.

That is unlikely to leave the hazard unchanged. Suppose, for illustration, that it rises to 3.38 per cent, the level SolaraPay's Tier 2 was recalibrated to. The eighteen month plan then collects about thirty one thousand five hundred and fifty two. That is less than the twenty four month plan.

Lengthening the tenor fails the other way. Over thirty months at the default hazard, the expected collected share is 60.0 per cent. The instalment needed is about one thousand eight hundred and seventy nine, a burden of 10.4 per cent. Still above the threshold, and with six more months of exposure.

Tenor alone cannot reconcile this product with this customer. The workbook holds the hazard constant across tenors, so you must decide whether a change in tenor or burden changes the hazard, and run that judgement through Products and Scenarios.

## Scene 6. Upgrades, add ons and warranty

*On screen: Repeat customers: lowest risk. Watch rolled balances. Warranty: a fault becomes a credit loss. Battery life against tenor.*

A customer who completes a plan is the lowest risk prospect the company will ever have. But an upgrade that rolls an unpaid balance into a new, larger contract can extend the tenor and hide arrears. Ask for upgrade cohorts to be reported separately.

Warranty is part of the price plan. A customer whose system fails stops paying, so a slow warranty turns a hardware fault into a credit loss. For Tiers 4 and 5, check that battery life and warranty cover the thirty six or forty eight month tenor, and that supplier warranties match the warranty given to customers.

## Scene 7. Product mix and untested tiers

*On screen: Tier 4 and 5 mix 20% and 10%: IRR 46.4% to 66.6%. SolaraPay Tier 4 and 5 hazard doubled: IRR 29.0% to 14.8%, multiple 3.6 to 2.0 times, 38 breach months.*

Higher tiers look better per unit. In the workbook they carry lower hazards, larger contracts and higher advance rates, 75 per cent for Tiers 4 and 5 against 50 per cent for Tier 1. Raising the Tier 4 and 5 mix to 20 and 10 per cent lifts the default investor internal rate of return, or IRR, from 46.4 to 66.6 per cent, with no covenant breach.

The difficulty is that those assumptions are, for most companies, untested. Shifting the mix towards them is the largest credit bet in the plan, made on the least evidence.

SolaraPay is launching Tier 4 at three hundred and ninety thousand shillings and Tier 5 at eight hundred and forty five thousand. Calibrated unit economics give them the highest ratios of lifetime value to acquisition cost in the range, 11.0 and 14.9 times. But they have no history. Doubling their hazard cuts the investor IRR in the calibrated case from 29.0 to 14.8 per cent. The multiple falls from 3.6 to 2.0 times, with 38 months in covenant breach. The case caps both tiers as a pilot, released only after twelve months of cohort data within 10 per cent of plan.

## Scene 8. What SolaraPay's Tier 2 shows

*On screen: SolaraPay Tier 2. Expected loss 41.8%. Unit IRR 37%. Payback 17 months. 40% of mix. Burden 13.0%. Calibrated collected share 58.2%.*

Tier 2 is SolaraPay's weakest product on the measures a lender cares about most. Its expected loss is 41.8 per cent, the highest in the range. Its unit IRR is 37 per cent, the lowest. It pays back in seventeen months on a twenty four month plan. Its lifetime contribution of seven thousand five hundred and twenty four shillings is 19.3 per cent of its cash price. And it is the largest line in the plan, at 40 per cent of the mix.

Its burden is 13.0 per cent, above the threshold. Calibration raised its hazard from 2.60 to 3.38 per cent and cut its collection rate from 88 to 87 per cent. On those parameters, the expected collected share over twenty four months falls from 64.4 to 58.2 per cent.

The response is redesign, not withdrawal. Bring the instalment within the threshold, through a smaller product, a lower premium, a higher deposit for some segments or a different target customer. Then test the redesign on new cohorts before scaling it. New prices also need indexing to the exchange rate. In the default sensitivities, freezing prices on new contracts cuts the Base investor IRR from 46.4 to 7.9 per cent.

## Scene 9. Recap and exercise

On screen: *Exercise. Tier 1 defaults: hazard 3.5%, collection 88%, tenor 12 months. Expected collected share. Compare with Unit\_Economics and D2.*

To recap. The four price plan inputs move revenue, exposure, affordability and risk together. Expected collection falls with tenor, and neither a shorter nor a longer tenor alone fixes an unaffordable plan. Untested tiers deserve a pilot cap and a doubled hazard stress.

Your exercise applies the tenor mechanics to Tier 1. Use the default monthly hazard of 3.5 per cent and the collection rate of 88 per cent. Compute the expected collected share over a twelve month tenor, then over eighteen months. Check your answers in the D3 repayment curve tool or the D2 unit economics calculator. Then decide, in one paragraph, whether a longer Tier 1 tenor would be worth its extra exposure.

## Module 5. Distribution and customer acquisition

Running time: 9:35. Book: Chapter 5. Model sheets: Products, Inputs, Unit\_Economics, Vintage\_Input. Templates and tools: T04 Agent economics, D2 Unit economics calculator.

### Learning objectives

1. Build customer acquisition cost on a narrow and a fully loaded definition, and reconcile the two to the management accounts.
2. Explain why commission design and agent quality are credit variables, and read vintage curves cut by agent, channel and territory.
3. Recognise the data traces of fraud and ghost sales, and test acquisition cost together with credit in the unit economics.

### Scene 1. A customer is a credit decision

On screen: *Title "What does a customer really cost?" with the line "A cheap customer who defaults in month four is the most expensive one".*

The decision is what a customer really costs to acquire, and whether that cost is recovered before the customer stops paying.

In a PAYGo company, PAYGo meaning pay as you go, that is a credit question. The people who find customers also decide, in practice, who gets credit. A cheap customer who defaults in month four is the most expensive customer the company will ever sign.

### Scene 2. Who finds the customer

On screen: *Three channels: direct sales agents, retail partners, installers.*

PAYGo solar is sold through people. The typical customer is a rural household with a mobile money wallet and a seasonal income.

Three channels carry most of the work. Direct sales agents are the backbone, often paid entirely on commission. The company controls them only through the commission plan, the training and the data it watches.

Retail partners, such as agro dealers and phone shops, bring footfall. Their incentive is the sale, not the repayment.

Installers matter most for Tiers 3 to 5. When a customer says the system does not work, that often means, in collections language, that they are not going to pay. So installer quality shows up in early default rates as well as warranty costs.

### Scene 3. Commission structures

On screen: *Flat commission, split commission, portfolio bonus, clawback.*

The simplest plan pays a flat commission when the deposit is received and the system is activated. It is also the plan most likely to fill the book with accounts that never make a second payment.

The alternatives split the commission. Part is paid on activation and part later, if the account is still current at a checkpoint such as month three. The most direct form is the clawback, where commission already paid is recovered if the account defaults within a set window.

Each design has a cost. Deferred pay raises attrition among new agents. And where agents can simply leave for a competitor, a large clawback is a deterrent on paper only.

Here is a practical test for an investment committee. Ask what share of commission paid in the last twelve months was contingent on repayment. Then ask how much of it was actually withheld. If the second answer is close to zero, the contingency is not working.

#### Scene 4. Fully loaded acquisition cost

*On screen: Table from the book: narrow CAC of 3,260 per sale against fully loaded CAC of 6,060 per sale, Tier 2 territory, 300 sales a month.*

The workbook defines customer acquisition cost, or CAC, as commission plus marketing per unit sold, entered by tier on the Products sheet. Most plans understate the true cost, because the field organisation sits in fixed overheads.

Take an illustrative territory selling a Tier 2 system, with figures in the local currency of the model's fictional market. Fifty agents each make six sales a month, so three hundred sales. Upfront commission is one thousand eight hundred local currency units a sale. A repayment linked commission of one thousand two hundred is paid on the 80 per cent of accounts still current at month three, which is nine hundred and sixty a sale. Marketing adds five hundred. The narrow CAC is three thousand two hundred and sixty a sale.

Now add the costs that exist only because the company is acquiring customers. Field supervisors add one thousand five hundred a sale. Recruitment and training to replace leavers add four hundred. Transport and fuel add seven hundred. Verification calls and fraud losses add two hundred. The fully loaded CAC is six thousand and sixty a sale, 1.86 times the narrow figure. On a cash price of thirty six thousand, that is about 16.8 per cent of the price, against 9.1 per cent on the narrow view.

Neither figure is wrong, as long as everyone knows which one is on the page. Ask for the reconciliation from the per unit figure to total sales and marketing expense in the management accounts. The gap is usually where the field organisation lives.

#### Scene 5. Agent quality as a credit variable

*On screen: Two cumulative repayment curves for the same product: top half of agents at 76 per cent, bottom half at 62 per cent at month 12.*

Two agents in the same territory, selling the same product in the same month, can originate books that repay very differently.

Take a Tier 2 contract with an instalment of two thousand local currency units a month. By month twelve, twenty four thousand has fallen due. Suppose accounts from the top half of agents have repaid 76 per cent of that, and accounts from the bottom half 62 per cent. The gap of fourteen points is worth three thousand three hundred and sixty per account by month twelve alone. That is more than the entire narrow CAC, and it keeps growing in the second year.

So agent identity belongs in the cohort data. A credit officer should see vintage curves cut by agent, agent tenure, channel and territory. Full curves take two years to form, but early signals arrive within a quarter: deposit only accounts, accounts more than thirty days past due at month three, and systems never active for a full month.

### Scene 6. Fraud and ghost sales

On screen: *Warning signs: shared phone numbers, shared wallets, month end bursts, devices never switched on.*

Any plan that pays per activation invites fabricated activations. An agent registers a fictitious customer, pays the deposit from their own wallet and earns a commission larger than the deposit.

Most of this leaves traces. Deposit only accounts cluster by agent. Several accounts share a phone number or a wallet. Installations arrive in bursts just before the commission cut off. Some devices are never switched on.

The controls are routine: verification calls, location tagged installation photographs, separate deposit and commission wallets, and random field audits. Their cost belongs in fully loaded CAC. A company that shows no fraud losses at all usually has not looked.

For a lender, ghost sales are worse than ordinary defaults. A fictitious account looks current in its first month and counts as collateral. Eligibility criteria should exclude deposit only accounts and accounts without a first instalment.

### Scene 7. Opening new territories

On screen: *Fixed cost per sale in a new territory: 9,000 at 100 sales, 4,500 at 200 sales, 3,000 at 300 sales.*

New territories are expensive before they are productive. Suppose a new territory carries fixed costs of nine hundred thousand local currency units a month. At maturity it sells three hundred units a month, so those costs add three thousand a sale. In the first month, at one hundred sales, they add nine thousand. At two hundred sales they add four thousand five hundred.

Early cohorts in a new area also tend to repay worse. A plan that applies mature hazard rates to a new territory is optimistic twice, on cost and on credit. Treat it as a pilot, with a budget, a ramp and stop criteria.

### Scene 8. How CAC enters the unit economics

On screen: *Unit\_Economics sheet. SolaraPay LTV to CAC by tier: 2.4, 3.5, 5.5, 11.0 and 14.9 times.*

On Unit\_Economics, CAC is paid in full on the day of sale and recovered, if at all, in small daily payments. That timing is why it weighs so heavily on payback.

In the calibrated SolaraPay case, a fictional company, the ratio of lifetime value to CAC, or LTV to CAC, runs from 2.4 times for Tier 1 to 14.9 times for Tier 5. Cash payback runs from 9 to 22 months. The large systems look far better because acquisition cost does not rise with contract value. They also carry the least evidence, since Tiers 4 and 5 have no repayment history.

Payback must be read against survival. For Tier 2 at the default monthly hazard of 2.6 per cent, about 27 per cent of accounts have stopped paying by month twelve, and about 38 per cent by month eighteen. A Tier 2 payback of 17 months is recovered, on average, from the customers still paying.

## Scene 9. Recap and exercise

On screen: *Template T04 Agent economics: narrow and fully loaded CAC, agent ranking, new territory sheet.*

Acquisition cost is a credit variable, because the people who sell also select the borrowers. Measure CAC on both definitions and reconcile it to the accounts. Read vintage curves by agent and channel, and price fraud controls into the cost. Then test CAC and credit together in the unit economics.

Your exercise: open template T04 and enter the Tier 2 territory from this module. Confirm the fully loaded CAC of six thousand and sixty. Then cut the share of accounts qualifying for the deferred commission from 80 to 60 per cent and note how CAC per sale moves. In the workbook, move field costs from Inputs into the per unit acquisition line on Products, remove them from fixed costs, and check on Unit\_Economics how LTV to CAC and payback change for each tier.

# Module 6. Repayment behaviour: cohorts, curves and defaults

Running time: 9:50. Book: Chapter 6. Model sheets: Products, Curves, Cohort\_T1 to Cohort\_T5, Credit\_Assumptions, Credit\_Input, Vintage\_Input, Vintage\_Dashboard, Credit\_Portfolio, Checks. Templates and tools: D3 Repayment curve and cohort calibration, T06 Loan tape and data request.

## Learning objectives

1. Compute survival and cumulative repayment from a monthly default hazard and a collection rate, and read a cohort against plan.
2. Use DPD buckets, roll rates and cures, and explain why portfolio ratios improve in a growing book without any change in credit quality.
3. Calibrate hazards and collection rates to observed cohorts, and keep history and projection apart with Proxy and Actual modes.

## Scene 1. The assumption behind every number

On screen: *Title "How will customers actually pay?" with the chain: repayment, receivables, borrowing base, covenants, valuation.*

Revenue, receivables, the borrowing base, covenant headroom and valuation all flow from a small set of repayment assumptions. When those assumptions are wrong, the error rarely shows in the first year, because new sales hide it.

Cohorts, survival curves, arrears buckets and roll rates are how a credit officer finds the error before the lender's covenant test does.

## Scene 2. Cohorts and vintages

On screen: *A cohort grid: months of sale down the side, account age across the top.*

A cohort, or vintage, is the group of accounts sold in the same month for the same product. Customers behave according to how long they have had the product, not according to the calendar. So each cohort is measured against account age, in months since sale.

The workbook is built the same way. The Curves sheet defines how one unit of each tier behaves at each age. The cohort sheets, Cohort\_T1 to Cohort\_T5, apply those curves to every monthly cohort. The Ops sheet adds them up by calendar month. Portfolio figures are always sums of cohort behaviour, never assumptions in their own right.

## Scene 3. The survival model

On screen: *Curves sheet. Tier 2 survival: 72.9 per cent at month 12, 53.1 per cent at month 24.*

The repayment engine rests on one assumption. Each month, a fixed share of the accounts still paying stops paying for good. That share is the monthly default hazard.

Survival to a given age is one less the hazard, raised to the power of the age in months. The instalment collected at that age is the instalment times the collection rate on paying accounts times survival. The collection rate captures payers who skip days. The hazard captures customers who leave altogether.

Take the default Tier 2 product, an SHS, or solar home system, with a TV, sold over 24 months, with a hazard of 2.6 per cent and a collection rate of 88 per cent. At month twelve, 72.9 per cent of accounts survive. At month twenty four, 53.1 per cent. The expected payment in the last month is 46.8 per cent of the instalment due.

Tenor and hazard interact. Tier 1, with a hazard of 3.5 per cent over 12 months, ends its tenor with 65.2 per cent still paying. Tier 3, with 1.9 per cent over 30 months, ends with 56.2 per cent. A lower hazard over a longer tenor can leave a similar share of non payers. Tenor extension is a credit decision, not only an affordability one.

Real hazards are usually higher in the first months and lower later. A constant hazard calibrated at month twelve will understate early losses and overstate late ones, which matters for the timing of covenant tests.

#### Scene 4. Cumulative repayment curves

On screen: *Chart of Tier 2 cumulative repayment: 83.5, 80.3, 74.5, 69.2 and 64.4 per cent at months 3, 6, 12, 18 and 24.*

The most useful single picture of a cohort is its cumulative repayment curve. It is cumulative collections divided by cumulative instalments due, at each account age. Deposits are excluded from both sides, because every customer pays a deposit and it would flatter the ratio.

For default Tier 2, the curve reads 83.5 per cent at month three, 74.5 per cent at month twelve and 64.4 per cent at month twenty four. It falls throughout, because each month adds a full instalment due and a collection from a shrinking pool of payers.

The curve lets you compare each cohort with plan from its first month. A cohort five points below plan at month twelve will not close that gap by month twenty four unless something changes in collections. On a constant hazard, the gap widens.

#### Scene 5. Arrears buckets, roll rates and cures

On screen: *Buckets: current, 1 to 30, 31 to 60, 61 to 90, 91 to 180, over 180 DPD. Roll rate path: 10, 40, 60, 75 per cent.*

Day to day credit management works with arrears, measured in DPD, or days past due. The workbook uses six buckets, from current to over 180 days.

DPD needs a definition first. Some platforms count days since the device last had credit. Others measure the shortfall against the payment schedule. The schedule based measure reconciles to the receivable, so it is closer to how a lender thinks. Whichever is used, state it and keep it.

Roll rates measure the share of a bucket that moves to the next one a month later. In an illustration, 10 per cent of current accounts move to 1 to 30 days, 40 per cent of those roll on, then 60 per cent, then 75 per cent. Multiply the four and 1.8 per cent of a current pool reaches 91 days past due three months later.

A cure is an account that returns to current. PAYGo, or pay as you go, has more cures than most consumer lending, because the lights go off. But roll rates fall steeply past 60 or 90 days. An account dark for three months has usually found an alternative. The workbook's proxy engine assumes no cures in the cash forecast, and uses a cure rate of 30 per cent only in the indicative expected credit loss.

## Scene 6. Lockout and the default definition

On screen: *Default definition: 180 DPD in the workbook. Same definition for default flag, write off, repossession and lender data.*

The lockout loses its force once the customer stops valuing the service, for example when the battery degrades or the grid arrives. Then the account moves quickly through the buckets and stays there.

The workbook defines default at 180 days past due. Ninety days is another common choice. A longer definition delays recognition and lifts reported cures. A shorter one may count as defaulted customers who would have paid. The same definition should drive the default flag, the write off, the repossession trigger and the lender's data.

## Scene 7. Seasoning and the denominator effect

On screen: *Same customers, three growth rates: 64.4, 68.3 and 71.6 per cent portfolio collection rate.*

A new cohort's losses arrive over the next 12 to 30 months. A portfolio ratio is therefore a weighted average of cohorts at different ages, and growth sets the weights.

Take the Tier 2 curve and a company selling the same volume every month for two years. Its portfolio collection rate is 64.4 per cent. If sales instead grow 5 per cent a month, the same customers produce 68.3 per cent. At 10 per cent growth, 71.6 per cent. Nothing about credit has changed. Only the denominator has.

When growth slows, the ratios deteriorate even if behaviour is stable. In the workbook's Base case the collection rate falls from 84.5 per cent in Year 1 to 73.3 per cent in Year 5, with no change in hazards. Read cohort curves, and use lagged portfolio ratios, to see through it.

## Scene 8. Calibrating against observed cohorts

On screen: *SolaraPay month 12 repayment, observed against plan: Tier 1 64.6 against 70.3, Tier 2 69.4 against 74.5, Tier 3 76.3 against 79.6. Vintage\_Dashboard.*

The default curves are proxies for a fictional market. Once a company has twelve months of cohort history, they should be replaced.

The SolaraPay case, a fictional company with 24 months of synthetic history, shows the method. At month twelve every tier sat below plan, by 5.7, 5.1 and 3.3 points for Tiers 1 to 3. Recalibration set each hazard at 1.3 times its plan value. Tier 2 moved from 2.6 to 3.38 per cent, and its collection rate from 88 to 87 per cent. The refitted Tier 2 curve gives 70.1 per cent at month twelve, against 69.4 observed. Over the full tenor the effect is larger. Survival at month twenty four falls to 43.8 per cent, against 53.1 on the default curve.

Two cautions apply. One checkpoint can be matched by many pairs of hazard and collection rate, so use months six and eighteen to choose. And Tiers 4 and 5 had no history and kept proxy values, which is ignorance, not comfort.

In the workbook, set Credit\_Assumptions to Actual mode, load history into Credit\_Input and Vintage\_Input, and compare on Vintage\_Dashboard. History is reported as history, and projections stay on the calibrated curves.

## Scene 9. Recap and exercise

On screen: *Decision tool D3: curve sheet, calibration sheet, portfolio effect sheet.*

Every portfolio figure is a sum of cohort behaviour, so start from cohorts measured by account age. Survival and cumulative repayment follow from a hazard and a collection rate. Growth flatters portfolio ratios, and cohort curves remove that flattery. Calibrate to observed history before trusting any plan.

Your exercise: open decision tool D3 and enter the SolaraPay Tier 2 observation of 69.4 per cent at month twelve, with the matching points at months three, six and eighteen if your data hold them. Compare your fitted values with the case's 3.38 per cent hazard and 87 per cent collection rate. Then use the portfolio effect sheet to reproduce the 64.4, 68.3 and 71.6 per cent figures at zero, 5 and 10 per cent monthly growth.

## Module 7. Portfolio KPIs and PAYGo PERFORM

Running time: 11:28. Book: Chapter 7. Model sheets: PERFORM\_2026, Credit\_Portfolio, KPIs, Covenants, Vintage\_Dashboard, Credit\_Input, Vintage\_Input, Checks. Templates and tools: T05 Lender KPI report, T06 Loan tape and data request.

### Learning objectives

1. Define the five PAYGo PERFORM 2026 KPIs and the operational and lender metrics (collection rate, receivables at risk, PAR30 and PAR90, write offs), and state what sits in each denominator.
2. Explain how growth distorts portfolio ratios, and use lagged and vintage measures to correct for it.
3. Design a monthly lender dashboard that reconciles DPD buckets to gross receivables and reports covenant headroom.

### Scene 1. Is the portfolio healthy

On screen: Title "Is my portfolio healthy?" with three words: definition, denominator, growth.

A board, an investor and a lender all take the same decision every month: whether the portfolio is healthy. The answer depends less on which ratios are reported than on how they are defined, what sits in the denominator and how fast the book is growing.

A portfolio can show improving ratios while its customers behave worse.

### Scene 2. Why a common framework matters

On screen: PAYGo PERFORM 2026: five KPIs on repayment and ownership, from company contract data. The 2021 guide is historical.

Before common frameworks took hold, PAYGo companies, PAYGo meaning pay as you go, often defined their own metrics. A collection rate might include deposits or not. Default might mean 60, 90 or 180 days without payment.

PAYGo PERFORM is the industry's answer. Its first technical guide, published in July 2021 by CGAP, GOGILA and IFC Lighting Global, set out portfolio indicators such as the collection rate, receivables at risk and the write off ratio. That guide is now historical. The current standard is the PAYGo PERFORM KPIs technical guide published by GOGILA in June 2026. It narrows the standard to repayment and ownership, with five key performance indicators, or KPIs. They are the repayment rate paid versus plan, the repayment rate paid versus financed, paid versus plan at 90 days, paid versus plan at twice the contract term, and the ownership rate at twice the contract term.

These KPIs are calculated on contract data, day by day, and they exclude deposits and subsidies. They come only from the company's own data. The workbook is a planning model with monthly cohorts, so it does not compute them. Its PERFORM\_2026 sheet holds them as the company reports them, adds up numerators and denominators as the standard requires, and shows the model's approximations beside them, labelled as such. The older ratios remain useful as operational and lender metrics, each with its definition printed beside it, but never in place of a repayment rate.

### Scene 3. Collection rate and receivables at risk

On screen: *Collection rate example: 74 collected against 100 due gives 74.0 per cent. Adding deposits of 12 gives a misleading 86.0 per cent.*

The operational collection rate, which is not a PERFORM KPI, is cash collected against scheduled instalments, divided by the instalments that fell due in the period. Deposits are excluded from both sides. Take a month in which instalments due are one hundred million local currency units, collections seventy four million and deposits twelve million. The collection rate is 74.0 per cent. Adding deposits to the top line alone would show 86.0 per cent, a figure that rises whenever the company simply sells more.

Receivables at risk measures the share of the book sitting on accounts that have stopped paying or are seriously late. The workbook computes it from the unit curve, as the carrying amount of accounts that have left the paying pool, divided by gross receivables. Operational definitions usually apply a lateness threshold, commonly 30 days, to the full outstanding balance of the account. The workbook's default covenant caps receivables at risk at 15 per cent.

PAR30 and PAR90, portfolio at risk over 30 and over 90 days, are the outstanding balance of accounts more than 30 or 90 days past due, divided by gross receivables. The whole balance of a late account counts, not only the arrears.

### Scene 4. Write offs, repayment and ownership

On screen: *Write off ratio, cohort repayment ratio, ownership rate at twice the term from company data, active ratio, enabled rate.*

The write off ratio is the amount written off divided by average gross receivables, annualised. A company that writes off at 180 days past due shows lower write offs, and higher PAR90, than one that writes off at 90 days with the same customers.

The workbook's Vintage\_Dashboard reports a cohort repayment ratio: cumulative collections over cumulative instalments due, at a given account age. It follows the logic of the PERFORM repayment rate paid versus plan, but it is not a PERFORM calculation, because it is monthly and does not allocate payments to instalments. The ownership rate is the share of contracts fully paid by twice the contract term, so a 24 month contract is measured at month 48. Most companies cannot yet evidence it. The workbook accepts it only from company data and will not derive it from its own curves.

The active ratio and the enabled rate complete the set. Neither has one accepted definition, so print the definition beside the figure.

### Scene 5. Growth flatters the ratios

On screen: *PAR30 falls from 20.0 to 14.3 per cent after new sales, while lagged PAR30 reads 21.5 per cent.*

Portfolio metrics tell a lender how much of today's collateral is performing. Vintage metrics, by cohort and account age, tell an analyst whether recent customers behave better or worse than earlier ones.

Portfolio metrics are distorted by growth. Take an illustrative seasoned book of eight hundred million local currency units, with PAR30 of 20 per cent, so one hundred and sixty million at risk. Next quarter the company adds four hundred million of new receivables, of which 3 per cent, twelve million, is already more than 30 days late. Hold the old book constant. Reported PAR30 falls to 14.3 per cent. No customer behaved better. The denominator grew faster than arrears can form.

A lagged ratio corrects much of this. Divide today's late balance by gross receivables three months earlier gives 21.5 per cent, close to the seasoned book's true condition. The same effect runs in reverse as growth slows. In the workbook's Base case the collection rate falls from 84.5 per cent in Year 1 to 73.3 per cent in Year 5, with no change in assumptions. Covenants set on a fast growing borrower's Year 1 ratios are set on the most flattering numbers it will ever produce.

### Scene 6. A monthly dashboard for lenders

On screen: *Dashboard blocks: book, performance, risk, losses, vintage, facility, covenants, integrity. Sheets: Credit\_Portfolio, KPIs, Covenants, Vintage\_Dashboard, Checks.*

A lender report should let the reader judge four things without asking for more data: whether the collateral performs, whether newer cohorts are better or worse, the covenant headroom, and whether the numbers reconcile.

The pack fits on two pages. It shows the PERFORM 2026 KPIs once the company computes them on its contract data, gross receivables by tier, the collection rate for the month and the trailing three months, the days past due buckets, PAR30, PAR90 and lagged PAR30, write offs and recoveries, and the indicative expected credit loss.

The vintage block is the one most often missing, and it gives the earliest warning. It sets repayment by cohort against plan at months three, six and twelve. The covenant block should show headroom, not only pass or fail. The calibrated SolaraPay case, a fictional company, passes its 70 per cent trailing collection covenant with a lowest reading of exactly 70.0 per cent. That is a pass with no headroom.

### Scene 7. Reconciling buckets to gross receivables

On screen: *Bucket table in millions: current 700, 1 to 30 days 120, 31 to 60 days 50, 61 to 90 days 40, 91 to 180 days 60, over 180 days 30, total 1,000.*

The buckets must add up to gross receivables. Platform exports often omit written off accounts or report arrears instead of balances. The Checks sheet enforces the reconciliation in Actual mode.

Take an illustrative month end with one billion local currency units of gross receivables. Seven hundred million is current and one hundred and twenty million is 1 to 30 days late. The buckets beyond 30 days add to one hundred and eighty million, so PAR30 is 18.0 per cent. Beyond 90 days sits ninety million, so PAR90 is 9.0 per cent. Eligible receivables up to 30 days are eight hundred and twenty million. At a 70 per cent advance rate, the borrowing base is five hundred and seventy four million.

The table also tests the write off policy. A large over 180 bucket means defaulted balances are still on the books. Either policy is acceptable if disclosed, but not if it changes from month to month.

### Scene 8. Reading the SolaraPay history

On screen: *SolaraPay: collection 77.4 then 69.3 per cent, latest month 66.5 per cent, PAR30 17.0 per cent, ECL coverage 36.5 per cent. Sector collection rate about 62 per cent for 2021 to 2023, reported, pending the primary document.*

SolaraPay's 24 months of history are synthetic. The portfolio collection rate was 77.4 per cent in the first year and 69.3 per cent in the second. The latest month read 66.5 per cent, with PAR30 at 17.0 per cent and indicative ECL, or expected credit loss, coverage of 36.5 per cent.

Sector figures offer context only. The ESMAP and World Bank market trends report for 2024 is reported to give an average PAYGo collection rate of about 62 per cent for 2021 to 2023. Its page and definition are still to be checked against the report, and it is a collection rate, not a PERFORM repayment rate. Against that, 69.3 per cent might pass. Read with the cohort data, it shows customers repaying below plan in every tier with history, and a book whose ratios will drift as growth slows. The case responds with a collections plan and a 65 per cent trailing collection covenant with a cure period, not a covenant the company would breach in its first quarter.

The case also shows the gap impact funders care about: no ownership evidence yet. Paid off customers should be tracked now, so the evidence exists when cohorts reach the checkpoint.

### Scene 9. Recap and exercise

On screen: *Template T05 Lender KPI report: input, KPIs, covenants, vintage, dashboard.*

Ask for the PERFORM 2026 KPIs computed on contract data, and refuse a collection rate offered in their place. Print the definition beside every operational and lender metric. Exclude deposits from the collection rate. Read portfolio ratios beside lagged and vintage measures, because growth flatters them. Reconcile the buckets and report covenant headroom, not just compliance.

Your exercise: open template T05 and load the SolaraPay history. Confirm the latest collection rate of 66.5 per cent and PAR30 of 17.0 per cent. Then compare PAR30 with PAR30 lagged by three months, and mark every covenant whose headroom is under a tenth of its threshold as a watch item.

## Module 8. Revenue recognition and credit losses

Running time: 10:31. Book: Chapter 8. Model sheets: Cohort\_T1 to Cohort\_T5, FS, Annual, Credit\_Portfolio, Credit\_Assumptions, Products, Benchmark\_Compare, Calibration, Investment\_Readiness. Templates and tools: T01 Investment memo, T03 Customer credit policy.

### Learning objectives

1. Split a PAYGo contract into hardware revenue and financing income, and explain the judgements behind the split under IFRS 15.
2. Apply the IFRS 9 expected credit loss model to PAYGo receivables, including staging, the 30 and 90 day presumptions, write offs and repossession.
3. Follow one cohort through the workbook's simplified accounting, read ECL against financing revenue, and reconcile the management, auditor and lender views of profit.

### Scene 1. What profit is the company really making

On screen: *Title "What profit am I really making?" with two judgements: revenue split, expected credit loss.*

Every profit figure in a PAYGo plan, PAYGo meaning pay as you go, rests on two accounting judgements. The first is how much of each contract is a sale of hardware and how much is interest. The second is how much of the receivable the company expects never to collect, and when that expectation reaches the income statement.

Management, auditors and lenders often answer both differently, by enough to move a company from profit to loss.

### Scene 2. A sale with a financing component

On screen: *Contract value split: hardware revenue at cash price on installation, financing income over the tenor.*

Under IFRS 15, the International Financial Reporting Standard on revenue from contracts with customers, a PAYGo contract is usually a sale of a good with a significant financing component. The customer takes the device home on day one and pays more than the cash price over 12 to 48 months.

So the contract is split. Hardware revenue is recognised at the cash price when control passes, normally at installation. The difference between the total contract value and the cash price is financing income, recognised over the tenor. A company that books the whole contract on day one overstates revenue, and in a growing book it does so every year.

Judgements sit inside this. Title passes only on the final payment, but the common view is that the lockout protects the seller's security and control still passes on delivery. It is still a judgement. And the effective interest method front loads financing income compared with a straight line. Whether the practical expedient for contracts of one year or less applies to a 12 month Tier 1 contract is a matter to agree with the auditor.

### Scene 3. Expected credit losses under IFRS 9

On screen: *Stage 1: 12 month ECL. Stage 2: lifetime ECL after a significant increase in credit risk. Stage 3: credit impaired. Presumptions at 30 and 90 days past due.*

PAYGo receivables are held at amortised cost, and the allowance follows the IFRS 9 expected credit loss model, or ECL. Under the general approach, Stage 1 accounts carry a 12 month ECL. Stage 2 accounts, where credit risk has increased significantly since origination, carry a lifetime ECL. Stage 3 accounts are credit impaired, also at lifetime ECL.

Two rebuttable presumptions matter. A significant increase in credit risk is presumed beyond 30 days past due, and default no later than 90 days. Auditors tend to accept a rebuttal only with cohort evidence of cures.

There is a second debate. Some companies apply the simplified approach, with a lifetime allowance from day one and no staging. Others apply the general approach, treating the contracts as consumer loans in substance. Whatever the approach, ECL must reflect forward looking information, such as a drought or a currency shock.

#### Scene 4. Write offs and repossession

*On screen: Repossession rate by tier, from 0 per cent for Tier 1 to 70 per cent for Tier 5. SolaraPay observed loss given default about 99 per cent.*

IFRS 9 requires a write off when there is no reasonable expectation of recovery, without setting a number of days.

Repossession is where PAYGo differs from unsecured lending. A recovered device goes into inventory at no more than its net realisable value, the resale price less refurbishment and selling costs. In the workbook, net recovery is defaults after a lag, times the repossession rate, times the resale share, times the cash price, less a recovery cost of 15 per cent of the proceeds. Repossession rates rise from zero for Tier 1 to 70 per cent for Tier 5.

The evidence often disappoints. The SolaraPay case, a fictional company with synthetic history, shows an observed loss given default of about 99 per cent. A plan that relies on resale values for Tiers 4 and 5 before the company has repossessed and resold a meaningful number is relying on a hope.

#### Scene 5. How the workbook simplifies

*On screen: Workbook conventions: hardware at cash price, financing income straight line, lifetime loss charged at sale, missed instalments written off as they fall due.*

The companion workbook, MODEL 2, simplifies these rules for transparency. Hardware revenue is the cash price at sale. Financing income is the PAYGo premium divided by the tenor, straight line. The allowance is set up at origination equal to the cohort's lifetime expected missed instalments, and consumed as they are missed. Missed instalments are written off as they fall due.

The whole lifetime loss hits the income statement at the sale, which is conservative. Straight line income is lower early than effective interest, also conservative. And the workbook's gross receivables and write offs are not comparable with audited figures, which will differ. This is an analytical modelling treatment and does not constitute a determination of the applicable accounting treatment under IFRS.

#### Scene 6. One cohort, worked through

*On screen: Tier 2 cohort of 1,000 units. Year 1: revenue 44.00 million, ECL charge 17.10 million. Year 2: revenue 8.00 million, no charge.*

Take an illustrative Tier 2 cohort of one thousand units. Each has a cash price of thirty six thousand local currency units and a deposit of four thousand, so thirty two thousand is financed. The customer pays two thousand a month for 24 months. The contract is worth fifty two thousand, so the premium is sixteen thousand.

On the default curve, with a hazard of 2.6 per cent and a collection rate of 88 per cent, expected collections are thirty thousand eight hundred and ninety six per unit. Missed instalments are seventeen thousand one hundred and four, or 35.6 per cent of instalments due. The cohort's allowance at origination is seventeen point one million.

In Year 1 the cohort brings forty four million of revenue, thirty six million of hardware plus eight million of financing income, against the full seventeen point one million of credit cost. In Year 2 it brings eight million of revenue and no credit cost. A growing company's income statement always carries the full lifetime loss of the newest cohorts against only their first months of financing income.

Under the effective interest method, at about 3.53 per cent a month, financing income would be about eleven point three million in Year 1 and four point seven million in Year 2. The total is the same, the timing is not.

### Scene 7. Indicative ECL and the ratio to financing revenue

*On screen: Credit\_Portfolio sheet. Indicative ECL 323.6 million on 1,000 million of receivables, coverage 32.4 per cent. SolaraPay ECL to financing revenue 1.28 times. Peer reference withdrawn.*

The Credit\_Portfolio sheet also computes an indicative stage based ECL as a diagnostic. Take a book of one billion local currency units, with eight hundred and twenty million current or up to 30 days late, ninety million in Stage 2 and ninety million in Stage 3. With Tier 2 inputs, a 30 per cent cure rate and an assumed resale share of 40 per cent, the loss factors are 22.85, 59.01 and 92.35 per cent. The indicative ECL is three hundred and twenty three point six million, a coverage of 32.4 per cent. SolaraPay reports 36.5 per cent in its latest month.

The calibrated SolaraPay plan shows an ECL charge of 1.28 times its financing revenue. Earlier drafts set this against a reference drawn from reported M-KOPA group figures. That comparison has been withdrawn, because the group figures conflict and the group's consolidated accounts have not been read. So the ratio is read on its own terms: financing income that does not cover expected losses means the price of credit does not pay for its risk. Part of it is mechanical. One Tier 2 cohort on the workbook's conventions has a lifetime ratio of 1.07 times and a first year ratio of 2.14 times, and a growing book sits above the lifetime figure. Part is credit, and part is pricing. The ratio is a good question and a poor verdict. Net credit losses against revenue, 37.5 per cent in Year 1 falling to 27.5 per cent in Year 5, answer it more directly.

### Scene 8. Three views of profit

*On screen: Management view, auditor view, lender view, reconciled side by side.*

Management tends to prefer judgements that smooth profit: a 12 month allowance on current accounts, a default definition beyond 90 days, optimistic recoveries and effective interest. Together they move credit cost later and income earlier.

The auditor tests whether those judgements are supportable.

The lender relies on neither. Its borrowing base counts eligible receivables at an advance rate and ignores the allowance. Its covenants test collections and arrears, not ECL. Changing ECL assumptions does not move the borrowing base by a single unit of currency. Present all three views side by side and reconcile them.

## Scene 9. Recap and exercise

On screen: *Exercise: rebuild the Tier 2 cohort, then change the cure rate on Credit\_Assumptions and read Credit\_Portfolio.*

Split each contract into hardware revenue at the cash price and financing income over the tenor. Agree the IFRS 9 approach, the staging presumptions and recovery values with the auditor, and support them with cohort data. The workbook charges lifetime loss at sale, so growth weighs on reported profit.

Your exercise: rebuild the Tier 2 cohort table by hand and confirm the year end receivable of sixteen million against an allowance of about eleven million after Year 1. Then set the cure rate on Credit\_Assumptions from 30 per cent to zero, read the change in indicative coverage on Credit\_Portfolio, and write one paragraph for the investment memo template, T01, explaining why the lender's borrowing base did not move.

## Module 9. Unit economics

Running time: 9:50. Book: Chapter 9. Model sheets: Unit\_Economics, Consumer\_Risk, Products, Curves, Vintage\_Dashboard, Investment\_Summary. Templates and tools: D2 Unit economics calculator, D3 Repayment curve and cohort calibration, T01 Investment memo.

### Learning objectives

1. Build the expected lifetime cash flow of one PAYGo unit from its inflow and outflow lines.
2. Compute and interpret cash payback, unit IRR, unit NPV and LTV to CAC, and explain why LTV to CAC on its own misleads.
3. Read the SolaraPay unit table tier by tier and connect unit economics to overheads, growth and the cost of funding.

### Scene 1. A unit is not a sale

On screen: Title "A unit is a customer relationship". Three words: tenor, cash, probability.

This module supports one decision: whether each sale creates value once the cost of acquiring, equipping, financing and servicing the customer is set against the cash that customer will actually pay. If a single unit loses money, no amount of volume will rescue the company.

In a cash retailer, a unit is a sale, booked and collected on the same day. In a PAYGo, or pay as you go, company, a unit is a customer relationship that lasts the full tenor and sometimes longer. The income statement books the hardware margin on the day of sale. The cash arrives instalment by instalment, from a population that thins out each month as accounts lock out.

So unit economics is a cash exercise. The useful unit is the expected lifetime cash flow of one customer at origination. A unit is a probability weighted customer, not the median good payer, and the weights come from the cohort curves.

### Scene 2. The lines of the unit

On screen: *Unit\_Economics sheet. Inflows: customer cash, recoveries, RBF. Outflows: hardware, installation, warranty, CAC, servicing.*

The Unit\_Economics sheet of the companion workbook, MODEL 2, sets out the unit for each tier. Inflows are the deposit plus expected instalments collected, net recoveries from repossessed and resold systems, and any RBF, or results based financing, payment attached to the unit. Outflows are hardware landed cost, installation, a warranty provision, CAC, the customer acquisition cost of commission plus marketing, and servicing.

Two disciplines apply. Every line is per unit sold, not per surviving account, so servicing is scaled to expected active months, which are fewer than the tenor. And expected loss and recoveries must come from the same curve. Take one from a sober source and the other from an optimistic one, and you build a unit that never existed.

### Scene 3. An illustrative unit

On screen: *Illustrative unit build: deposit 3,000, instalments 24 times 1,500, expected loss 30 per cent, contribution 8,800 local currency units.*

Take an illustrative mid tier system with a television, in a fictional local currency. The deposit is three thousand local currency units, followed by 24 monthly instalments of one thousand five hundred. The expected loss rate is 30 per cent of contractual instalments.

Contractual instalments total thirty six thousand. Thirty per cent of that is lost, so expected collections are twenty five thousand two hundred. Add the deposit, recoveries of one thousand two hundred and an RBF payment of one thousand, and expected inflows are thirty thousand four hundred.

Against that, hardware costs fourteen thousand, installation one thousand, warranty seven hundred, acquisition three thousand five hundred and servicing two thousand four hundred. Outflows total twenty one thousand six hundred. Lifetime contribution is eight thousand eight hundred local currency units, about 23 per cent of contract value. It is also the most flattering number in the analysis, because it is undiscounted.

### Scene 4. Timing the cash

On screen: *Cumulative cash by half year: minus 16,200 at month 0, minus 1,900 at month 12, payback about month 14. Unit IRR about 48.8 per cent. NPV at 25 per cent about 3,280.*

At month 0 the company receives the deposit and pays for hardware, installation, warranty and acquisition. Net cash is a loss of sixteen thousand two hundred. Collections then decline as the survival curve falls, and recoveries arrive in the second year.

Three measures follow. Cash payback is where cumulative cash turns positive. At month 12 the unit is still one thousand nine hundred short, and payback comes at roughly month 14. For a 24 month product that is acceptable but not generous.

Unit IRR, the internal rate of return, is about 48.8 per cent a year. Unit NPV, the net present value at a 25 per cent hurdle, is about three thousand two hundred and eighty. That is a little over a third of the undiscounted contribution, and the gap is the first reason to treat contribution with care.

### Scene 5. LTV to CAC and its pitfalls

On screen: *LTV to CAC: 12,300 divided by 3,500 equals 3.5 times. Three weaknesses: undiscounted, before overheads, loss sensitive.*

Lifetime value to customer acquisition cost, LTV to CAC, is the most quoted unit metric in pitch decks and the least reliable in credit papers. Here lifetime value is contribution plus CAC, twelve thousand three hundred, and the ratio is 3.5 times. Definitions vary, so ask which numerator and denominator were used.

The ratio has three weaknesses. It is undiscounted, so a 48 month unit and a 12 month unit can look alike. It is struck before overheads, so a ratio of 3 times can sit alongside a loss making company. And it is highly sensitive to the loss assumption.

Raise the expected loss from 30 to 40 per cent. Contribution drops to five thousand two hundred, a fall of 41 per cent, and the ratio falls to 2.5 times. Payback slips to about month 17, the unit IRR to about 28 per cent, and the NPV to about four hundred and eighty. A ten point error in the loss rate takes the unit from comfortably value creating to marginal.

## Scene 6. The SolaraPay unit table

On screen: *Unit\_Economics, SolaraPay calibrated, Tiers 1 to 5: expected loss, contribution, LTV to CAC, payback, unit IRR. Consumer\_Risk burden flags on Tiers 2, 3 and 4.*

SolaraPay is the book's fictional company, in the fictional Republic of Kivara, with 24 months of synthetic history. Tiers 1 to 3 are recalibrated to that history. Tiers 4 and 5 are new and keep default assumptions. Every tier has positive contribution. That is a minimum condition and no more.

Tier 2 is the weakest where it matters: the highest expected loss at 41.8 per cent, the lowest unit IRR at 37 per cent, and a 17 month payback on a 24 month tenor. Its payment burden, the instalment as a share of illustrative household income, is 13.0 per cent against a 10 per cent threshold on Consumer\_Risk. Part of the loss is an affordability problem. Tier 2 is also 40 per cent of the planned mix, so its redesign should be a condition of investment.

Tier 1 has the lowest LTV to CAC, 2.4 times, and a contribution of one thousand and ninety two shillings, yet a 66 per cent IRR and a nine month payback. It returns a small investment quickly but creates little value, so judge it as an upgrade funnel.

Tiers 4 and 5 look strongest on paper, at 11.0 and 14.9 times. They are the least proven. Their hazards are untested, and their recoveries assume repossession rates of 60 and 70 per cent against an observed loss given default proxy of about 99 per cent. Doubling their hazards cuts the investor IRR from 29.0 to 14.8 per cent. Hence the pilot cap.

## Scene 7. From the unit to the company

On screen: *Three bridges: overheads, growth, cost of funding. Blended funding cost 18.7 per cent.*

Three bridges connect the unit to the company. The first is overheads. In the illustration, central costs of six hundred million local currency units a year need about 68,200 units just to cover them. Under the 40 per cent loss case, that rises to about 115,400.

The second is growth. Each unit needs sixteen thousand two hundred at origination and pays back only around month 14, so doubling sales doubles the investment before earlier units repay.

The third is funding. With 70 per cent of each receivable funded at 16 per cent and 30 per cent by equity expecting 25 per cent, the blended cost is 18.7 per cent. The unit at about 49 per cent clears it easily. At about 28 per cent, little is left for overheads.

At SolaraPay every tier contributes, yet the calibrated Base margin on earnings before interest, tax, depreciation and amortisation is minus 35.2 per cent in Year 1 and turns positive only in Year 3. The units are profitable. The company becomes so only once the book can absorb its overheads and its financing.

## Scene 8. Recap and exercise

On screen: *Recap: cash not sales, read metrics together, Tier 2 redesign, Tiers 4 and 5 pilot. Exercise: D2 Unit economics calculator.*

A PAYGo unit is a probability weighted customer, measured in cash over its whole life. Read contribution, payback, unit IRR, unit NPV and payment burden together, and never accept LTV to CAC alone. At SolaraPay, Tier 2 needs redesign, and Tiers 4 and 5 stay a pilot until twelve months of cohort data sit within 10 per cent of plan.

Open the D2 unit economics calculator on its default Tier 2 sale and note contribution, payback month and unit IRR. Raise the default hazard until the expected loss rate is ten points higher, and record the same figures. Then write two sentences for a memo on whether the unit still clears a blended funding cost of 18.7 per cent.

## Module 10. Working capital, inventory and FX

Running time: 9:34. Book: Chapter 10. Model sheets: Inputs, Scenarios, Products, FS, Annual, Credit\_Portfolio, Investment\_Summary, Sensitivity. Templates and tools: D4 Receivables financing calculator, T07 Borrowing base and term sheet, T01 Investment memo.

### Learning objectives

1. Explain why the receivables book, not fixed assets, is the main investment of a PAYGo company, and estimate how it grows with sales.
2. Quantify the three effects of a local currency depreciation and the role of FX pass through and price indexation.
3. Trace how the model builds the funding requirement from cash before facility, the borrowing base and the equity top up, and read the SolaraPay currency stress cases.

### Scene 1. The book is the investment

On screen: Title "The receivables book is the main investment". Income statement versus cash flow statement, side by side.

This module supports the decision that most often goes wrong in PAYGo, or pay as you go, solar: how much cash a growth plan will consume before it returns any, and in which currency.

A PAYGo company has few fixed assets. Its main investment is the receivables book. The cash price less the deposit is, in substance, a loan to the customer.

The income statement hides this, because hardware revenue is booked on the day of sale. The cash flow statement shows the opposite: more sales mean a larger increase in receivables, and operating cash flow falls as growth accelerates. A board that reads only the first will approve a plan the balance sheet cannot carry.

### Scene 2. An illustrative receivables build

On screen: Table, Years 1 to 5: units sold 10,000 to 50,000, year end receivables 180 to 1,200 million local currency units, increase in receivables peaking at 360 million in Year 4.

Take an illustrative product with a cash price of twenty seven thousand local currency units, a deposit of three thousand and a 24 month tenor. Each unit adds twenty four thousand of principal to the book, repaid in equal monthly amounts.

Sales rise from 10,000 units in Year 1 to 50,000 in Year 4, then stay flat. The increase in receivables grows every year while sales grow, reaching three hundred and sixty million local currency units in Year 4. In Year 5, when sales stop growing, the increase drops to ninety million and principal collected almost matches new lending. The company turns towards cash generation with unchanged unit economics.

As a rule of thumb, a fast growing 24 month book sits at roughly one year of origination. Longer tenors push the multiple higher, which is why a shift towards Tiers 4 and 5 raises the funding requirement before it raises cash generation.

### Scene 3. Who funds the book, and inventory

*On screen: Effective advance: 75 per cent times 80 per cent equals 60 per cent. Inventory: three months cover, sixty days supplier credit.*

Suppose a lender advances 75 per cent against eligible receivables, and 80 per cent of the book is eligible. The effective advance is 60 per cent. On a book of one billion one hundred and ten million local currency units, the facility funds at most six hundred and sixty six million. The other four hundred and forty four million must come from equity and retained cash. Equity scales with the book, not with the sales line, and rises again when credit slips and eligibility shrinks.

Inventory is smaller, but risky. At 50,000 units a year and a landed cost of fourteen thousand, purchases run at seven hundred million a year. Three months of cover ties up one hundred and seventy five million. Sixty days of supplier credit funds about one hundred and fifteen million. Inventory is bought ahead of sales and in hard currency, and supplier credit is the first funding to vanish in a stress. Ask to see the plan with supplier credit at zero.

### Scene 4. Two currencies, one balance sheet

*On screen: Exchange rate moves from 130 to 156 per dollar. Three effects: hardware cost, book value in dollars, dollar debt.*

Hardware and often term debt are in US dollars. Collections are in local currency, fixed for the whole tenor. When the dollar rises by 20 per cent, from 130 to 156, three things happen at once.

New hardware costs more. A unit landed at one hundred dollars rises from thirteen thousand to fifteen thousand six hundred local currency units.

The existing book loses value in dollars. One billion one hundred and ten million local currency units is worth about eight and a half million dollars at 130, and about seven point one million at 156. Signed contracts cannot be repriced.

Dollar debt gets dearer. A three million dollar term loan grows from three hundred and ninety million to four hundred and sixty eight million local currency units, a translation loss of seventy eight million through the income statement.

### Scene 5. FX pass through

*On screen: FX pass through 50 per cent: price 27,000 to 29,700, margin 14,000 to 14,100 local currency units, but 108 to 90 dollars.*

FX, or foreign exchange, pass through is the share of the currency move recovered by raising prices on new contracts. The model default is 50 per cent. With a 20 per cent rise in the dollar, the cash price of new contracts rises by 10 per cent, to twenty nine thousand seven hundred. The local currency margin per unit is roughly preserved, at about fourteen thousand. In dollars, it falls from about one hundred and eight to about ninety, a decline of about 16 per cent. And nothing has been done for the book already sold.

Higher pass through has a cost. Each price increase raises the instalment and the payment burden, and a tier already above the affordability threshold will feel it in arrears. Pass through is a pricing decision and a credit decision at once.

## Scene 6. What the sensitivities say

On screen: *Sensitivity sheet, investor IRR and breach months: Base 46.4 per cent and 0; depreciation 20 per cent a year, minus 16.0 per cent and 41; no price increase 7.9 per cent and 35; hardware cost up 15 per cent, 29.5 per cent and 38; volume down 20 per cent, 37.8 per cent and 0.*

The Sensitivity sheet runs one at a time tests from the default Base case, where the investor IRR, the internal rate of return, is 46.4 per cent with no covenant breaches. Cutting volume by a fifth costs about nine points of IRR and no breaches. Local currency depreciation of 20 per cent a year takes the IRR to minus 16.0 per cent, with the company in breach for 41 of 60 months. Freezing prices on new contracts takes it to 7.9 per cent. A 15 per cent rise in hardware cost gives 29.5 per cent and 38 breach months.

The ranking is the point. In this sector, currency and pricing are larger value drivers than volume. A Downside case that cuts volume but leaves the exchange rate on its Base path is testing the wrong thing.

## Scene 7. Cash before facility and the funding requirement

On screen: *Formula in words: drawn balance is the lower of limit, borrowing base and the balance needed to hold minimum cash. Example: borrowing base 820, shortfall 40 million, equity top up.*

The model builds the funding requirement in a fixed order. It first computes cash before facility, which is opening cash plus the month's operating, investing and term debt flows. It then draws the facility to hold cash at the minimum balance, within the lower of the limit and the borrowing base. Any shortfall is met by an automatic equity top up, and the cumulative top up is the funding requirement.

Say the facility starts the month at eight hundred million, minimum cash is one hundred million and cash before facility is forty million. The company wants to draw to eight hundred and sixty million. If the borrowing base has shrunk to eight hundred and twenty, cash closes at sixty million, and the forty million gap is an equity top up.

The facility limit is rarely the binding constraint. The borrowing base is, and it shrinks just when collections weaken. At default inputs, peak equity is ten million dollars in Base and forty six point six million in Severe.

## Scene 8. SolaraPay: currency decides

On screen: *SolaraPay stress table: Calibrated Base, Downside, Downside with full FX indexation, Severe. IRR, multiple, breach months.*

SolaraPay, the book's fictional company in the fictional Republic of Kivara, plans in shillings at 135 per dollar, with dollar hardware, a three million dollar term loan at 10 per cent and a proposed four point five billion shilling facility at 16 per cent. Its nine million dollars of initial equity is enough in the calibrated Base.

The calibrated Downside combines higher default hazard, lower collections, lower volume, dearer hardware and depreciation of 12 per cent a year. The investor loses everything. Change one input, so that new contract prices are fully indexed to the exchange rate, and the IRR recovers to 23.6 per cent with a multiple of 2.9 times.

Two cautions apply. Indexation rescues equity value, not covenants, with 38 breach months against 40. And it assumes customers accept higher prices without repaying worse. That is why the recommendation pairs indexation with an affordability redesign that keeps the instalment within 10 per cent of income.

## Scene 9. Recap and exercise

On screen: *Recap: book grows with sales, three currency effects, borrowing base binds. Exercise: D4 Receivables financing calculator.*

The receivables book absorbs cash in proportion to growth, and equity funds what the borrowing base will not. A weaker local currency raises hardware cost, shrinks the book in dollars and inflates dollar debt at once. Currency and pricing move returns more than volume, and for SolaraPay price indexation is the strongest Downside mitigant.

Open the D4 receivables financing calculator on its default Tier 2 plan and note peak funding, the facility's share and peak equity. Then, on its currency sheet, enter a 10 per cent dollar rate and a 16 per cent local rate, and read the depreciation at which the two cost the same. Compare it with the 12 per cent a year in the Downside, and write one line on which currency should fund the book.

# Module 11. Funding the book: equity, debt and receivables facilities

Running time: 9:49. Book: Chapter 11. Model sheets: Inputs, Products, Credit\_Assumptions, Credit\_Portfolio, Covenants, KPIs, Credit\_Input, Vintage\_Input, Sensitivity. Templates and tools: T07 Borrowing base and term sheet, T05 Lender KPI report, D4 Receivables financing calculator.

## Learning objectives

1. Match each layer of the funding stack to the evidence a PAYGo company can show at each stage, and explain the three jobs of equity.
2. Build a borrowing base from receivables by tier and DPD bucket, apply eligibility and a concentration limit, and measure how fast headroom disappears.
3. Judge which covenants test a growing PAYGo book well, and set out a lender's options on the SolaraPay facility.

## Scene 1. The funding stack by stage

On screen: *Four stages: launch, early growth, scale, mature. Matching sources: equity and grants, hard currency term debt, local currency facilities, securitisation.*

This module supports the decision of how growth should be financed: how much equity, which debt, in which currency and on what terms. The funding stack has to match the receivables book in size, tenor and currency.

Lenders lend against data. At launch, with a few cohorts, a PAYGo, or pay as you go, company raises equity, grants, results based financing and supplier credit. With 12 to 24 months of vintages it adds hard currency term debt from impact lenders. With full tenor cohorts and audited accounts it can raise local currency receivables facilities. Only with several years of vintages and a clean data tape can it sell receivables to a special purpose vehicle, or SPV.

A downturn can push a company back a stage. GOGLA reports off grid solar investment of about two hundred and ninety nine million dollars in 2024, down about 30 per cent, a figure from secondary reporting still to be checked against the primary source.

## Scene 2. What equity is for

On screen: *Three jobs of equity: early losses, the unfunded part of the book, first loss. Peak equity: default and SolaraPay.*

Equity does three jobs. It funds early losses. It funds the part of the book that lenders will not advance against. And it absorbs first loss on the whole balance sheet, which is what allows lenders to advance at all.

Founders tend to underestimate the third. A lender advancing 70 per cent relies on the other 30 per cent, and on the equity beyond it, to absorb losses first. When credit deteriorates, eligibility shrinks and the advance falls. The companion workbook meets that shortfall with an automatic equity top up. For SolaraPay, the book's fictional company, peak equity is nine million dollars in the calibrated Base and forty one point nine million in the calibrated Severe case. An investor should know which figure it is underwriting.

### Scene 3. Hard currency term debt

On screen: *SolaraPay term loan: three million dollars at 10 per cent. 405 million shillings at 135, 486 million at 162. Loss of 81 million shillings.*

Dollar term loans from impact lenders and development finance institutions are often the first debt a company raises. They are patient and need no borrowing base. Their weakness is currency.

SolaraPay's three million dollar loan at 10 per cent, with a 12 month grace period and 36 months of amortisation, is four hundred and five million shillings at the opening rate of 135 per dollar. If the rate moves to 162, a 20 per cent rise in the dollar, the principal is four hundred and eighty six million. The company has lost eighty one million shillings before repaying anything. Use such debt for early losses and inventory, not for the book, and move the book to local currency as soon as you can.

### Scene 4. How a receivables facility works

On screen: *Facility terms: advance rates 50, 70 and 75 per cent, eligibility to 30 DPD, concentration limits, cash sweep.*

A receivables facility is a revolving loan secured on PAYGo receivables, usually in local currency. The amount available each month is the borrowing base: advance rate times eligible receivables, by product, less deductions for concentration. The model's default advance rates are 50 per cent for Tier 1, 70 per cent for Tiers 2 and 3 and 75 per cent for Tiers 4 and 5.

Eligibility decides what counts at all. The model's default is up to 30 days past due, or DPD. Lenders add criteria such as a lockable device and a verified customer, each needing an auditable data field. Concentration limits stop the base drifting towards assets the lender knows least about. Under a cash sweep, collections pass through a controlled account, and cash above an agreed minimum repays the facility first.

### Scene 5. A worked borrowing base

On screen: *T07 borrowing base certificate. Gross book 2,950, eligible 2,500, base before limits 1,746, concentration deduction 135, base 1,611, drawn 1,500, headroom 111. Figures in millions of shillings.*

Take an illustrative book, in millions of shillings. The gross book is two thousand nine hundred and fifty. Receivables current or up to 30 days past due are eligible, two thousand five hundred in all, about 85 per cent. Applying the advance rate tier by tier gives a base of one thousand seven hundred and forty six.

Now a concentration limit. Tiers 4 and 5 have less than twelve months of history and may not exceed 12 per cent of the eligible pool, a cap of three hundred. They contribute four hundred and eighty, so one hundred and eighty is excess. At a 75 per cent advance, the deduction is one hundred and thirty five. The borrowing base is one thousand six hundred and eleven.

Against a facility limit of two thousand and a drawn balance of one thousand five hundred, headroom is one hundred and eleven. The limit is not the constraint. The base is. And the effective advance on the gross book is about 55 per cent, well below every headline rate.

## Scene 6. How quickly headroom disappears

On screen: *Migration of 100 million of Tier 2 receivables past 30 DPD. Headroom falls from 111 to 32.*

Suppose one hundred million of Tier 2 receivables slip past 30 days, and nothing else changes. Tier 2's contribution to the base falls by seventy. The eligible pool shrinks, so the concentration cap falls to two hundred and eighty eight, and the deduction on Tiers 4 and 5 rises to one hundred and forty four. The base falls to one thousand five hundred and thirty two, and headroom from one hundred and eleven to thirty two.

A migration of about 3.4 per cent of the book has removed more than 70 per cent of headroom. Another such month would breach the base and force repayment from collections that are themselves weakening.

## Scene 7. Covenants that fit a growing book

On screen: *Covenants sheet defaults: collection rate 70 per cent, receivables at risk 15 per cent, 30 plus DPD 25 per cent, 90 plus DPD 18 per cent, headroom zero, debt to equity 3.0 times, DSCR 1.20 times.*

The model's default covenants are a trailing three month collection rate of at least 70 per cent, receivables at risk of no more than 15 per cent, 30 plus DPD of no more than 25 per cent, 90 plus DPD of no more than 18 per cent, non negative headroom, debt to book equity of no more than 3.0 times, and an annual DSCR, the debt service coverage ratio, of at least 1.20 times. These are operational and lender metrics, not PAYGo PERFORM KPIs. The 2026 standard forbids presenting a collection rate as the repayment rate, so the facility documents should print the definition of every covenant ratio, rather than borrow the company's internal dashboard. The PERFORM 2026 repayment rate can be added as a reporting line once the company computes it on its own contract data.

DSCR is ill suited to a growing book. Lending to new customers runs through operating cash flow, so coverage falls when growth accelerates, whatever the book's quality. SolaraPay's calibrated Base sits below 1.20 times in Years 1 to 4 while every portfolio covenant holds. Portfolio covenants with a cash sweep work better. A cure period, often one or two test dates, should be long enough for a collections plan to show results. Equity cures suit leverage covenants, not collection covenants, because new equity does not change how customers pay.

## Scene 8. SolaraPay from the lender's side

On screen: *SolaraPay request: 4.5 billion shilling facility at 16 per cent. Lowest projected trailing collection rate 70.0 per cent. History: 77.4, then 69.3, latest month 66.5 per cent. PAR30 17.0 per cent.*

SolaraPay asks a local bank for a four point five billion shilling facility at 16 per cent, with the default advance rates and eligibility to 30 days. On the calibrated projection it passes every monthly covenant, but only just. Debt to book equity peaks at 2.51 times in Year 3, against a limit of 3.0 times.

The weak point is collections. The lowest projected trailing collection rate is exactly 70.0 per cent, so no headroom. The history is worse: 77.4 per cent in the first year, 69.3 per cent in the second, and 66.5 per cent in the latest month. PAR30, the share of the portfolio more than 30 days past due, is 17.0 per cent and rising.

A lender has two sound options. One is to set the covenant at a level the book can meet, such as 65 per cent on a trailing three month basis, with a cure period and a written collections plan. The other is to defer the facility until the trend turns. Lending at 70 per cent and waiving the breach in month one is not a third option.

## Scene 9. Recap and exercise

On screen: *Recap: equity carries first loss, the borrowing base binds, portfolio covenants over DSCR. Exercise: T07 borrowing base certificate.*

The funding stack follows the evidence, and equity carries the first loss that makes debt possible. The borrowing base binds, and small migrations past 30 days can erase headroom. Portfolio covenants test a growing book better than DSCR, and SolaraPay's collection covenant must be set where the book can actually perform.

Open the T07 borrowing base certificate on its worked example and confirm the base of one thousand six hundred and eleven and headroom of one hundred and eleven. Then move Tier 2 receivables past 30 days in steps of fifty million, and record the step at which headroom turns negative.

# Module 12. Securitisation and off balance sheet structures

Running time: 9:53. Book: Chapter 12. Model sheets: Inputs, Products, Credit\_Assumptions, Credit\_Portfolio, Sensitivity, Credit\_Input, Vintage\_Input, Investment\_Readiness. Templates and tools: T06 Loan tape and data request, T07 Borrowing base and term sheet, T02 Due diligence checklist.

## Learning objectives

1. Explain how a true sale to an SPV, tranching, overcollateralisation, excess spread and triggers protect noteholders, and why the servicer matters so much in PAYGo.
2. Compute credit enhancement and excess spread for an illustrative pool, and show how a small rise in losses removes the cushion.
3. Apply the readiness criteria, the cost logic and the accounting test to decide whether a securitisation is worth pursuing.

## Scene 1. What a securitisation changes

On screen: *Diagram: originator sells receivables to an SPV in a true sale. SPV issues notes. Collections pay the notes.*

This module supports a decision that is usually taken too early: when a securitisation is feasible, and whether it is worth the effort.

A receivables facility is a secured loan to the company. A securitisation separates the receivables from the company. The originator sells a pool of receivables to a special purpose vehicle, or SPV, in a true sale. The SPV funds the purchase by issuing notes, and collections on the pool pay those notes in order of priority.

Two legal properties carry the structure. A true sale takes the receivables out of the originator's estate, so its creditors cannot claim them in an insolvency. Bankruptcy remoteness means the SPV has no other business and limited permitted activities. Together they let investors analyse the pool rather than the company.

In PAYGo, or pay as you go, solar, that separation is never complete. Someone must run the payment platform, issue unlock codes, collect through mobile money and repossess devices. If the originator fails, the receivables still belong to the SPV, but someone has to keep collecting them.

## Scene 2. Tranching and credit enhancement

On screen: *Illustrative pool of 5,000 million local currency units: senior 3,000 at 14 per cent, mezzanine 750 at 20 per cent, first loss 1,250 retained. Senior subordination 40 per cent.*

The SPV issues notes in layers. Senior notes are paid first and absorb losses last. Mezzanine notes sit below them. A first loss tranche, usually kept by the originator, absorbs losses first.

Take an illustrative pool of five billion local currency units. Senior notes are three billion, or 60 per cent. Mezzanine notes are seven hundred and fifty million, or 15 per cent. The retained first loss is one billion two hundred and fifty million, or 25 per cent. The senior notes are protected by the 40 per cent of the pool that ranks below them. Where notes issued are less than the pool, the difference is overcollateralisation, which works the same way.

A first loss of that size fits PAYGo. In the calibrated SolaraPay case, lifetime expected losses run from about 27 to 42 per cent by tier. A senior tranche needs protection against a multiple of expected loss, not just the expected figure.

### Scene 3. Excess spread

*On screen: One year snapshot: gross yield 2,250, losses 1,000, servicing 400, senior interest 420, mezzanine interest 150, other costs 50. Excess spread 230, or 4.6 per cent.*

The second line of defence is excess spread: pool income less losses, fees and note interest. This one year snapshot is stylised.

A gross yield of 45 per cent on five billion brings in two billion two hundred and fifty million. Credit losses at 20 per cent take one billion. Servicing at 8 per cent takes four hundred million. Senior interest is four hundred and twenty million, mezzanine interest one hundred and fifty million, and trustee and other costs fifty million. Excess spread is two hundred and thirty million, or 4.6 per cent of the pool.

That cushion is thin. If losses rise from 20 to 25 per cent, they take a further two hundred and fifty million, and excess spread becomes a loss of twenty million. A five point rise in losses, well within the gap between a management plan and a calibrated case, wipes out the cushion and starts eating into the first loss. PAYGo pools yield a lot because they lose a lot, and the margin between the two is what a structurer protects.

### Scene 4. Triggers, servicer and data tape

*On screen: Triggers: collection rate, PAR30, cumulative defaults, excess spread, servicer events. Backup servicer: cold or warm. T06 loan tape.*

Triggers change the flow of cash when performance slips. The most important is early amortisation: the SPV stops buying new receivables, stops releasing excess spread and applies all collections to the senior notes. Typical metrics are the trailing collection rate, PAR30, the share of the pool more than 30 days past due, cumulative defaults by vintage, and servicer events. Levels must come from the company's own history. A collection trigger at a level the book has already breached, as SolaraPay's history shows for a 70 per cent covenant, would amortise the deal on day one.

The servicer is the originator, because only it runs the platform and the agents. A backup servicer is appointed at closing. A cold backup holds the contract and would need months. A warm backup receives regular data and could take over in weeks. The test is simple: a backup that cannot keep devices unlocked for paying customers cannot protect collections.

Investors analyse the pool through a loan level data tape. It must cover vintages that have reached full tenor, reconcile monthly to the audited accounts, and print every definition and keep it stable over time, with repayment and ownership rates following the PAYGo PERFORM 2026 standard.

### Scene 5. Local currency issuance and the legal questions

*On screen: Reported transactions, with caveats: Sun King, Kenya, 2023 and 2025. d.light, five facilities since 2020. Legal checklist: true sale, assignment, SPV tax, data, licences.*

The main attraction is local currency funding at scale, which removes the translation risk of dollar debt. Local pension funds, insurers and banks may buy, if their rules allow.

The reported transactions give a sense of scale, from company and secondary reporting. Sun King is reported to have completed local currency securitisations in Kenya of about one hundred and thirty million dollars in May 2023, and about one hundred and fifty six million dollars, or twenty point one billion Kenyan shillings, in July 2025, arranged with Citi. d.light is reported to have five securitisation facilities since 2020 with about seven hundred and eighteen million dollars of purchasing capacity, which is capacity, not debt raised. These are large originators with long histories. They show the structure can work, not the terms a smaller company could expect.

Feasibility is first a legal question, to be confirmed with local counsel. Ask whether the law recognises a true sale of future instalments, whether receivables can be assigned without each customer's consent, whether the SPV is tax neutral, whether data and mobile money rules allow the transfer, and whether a lending licence is needed.

## Scene 6. Cost, size and the accounting

*On screen: Fixed costs of one million dollars over a three year life: 20 million dollar deal keeps 0.33 points of a 2 point saving, 100 million dollar deal keeps 1.67 points. Off balance sheet: usually not.*

Securitisations carry large fixed costs, from lawyers and arrangers to the tape review and months of management time. Suppose fixed costs are one million dollars and the notes have a three year average life. On a twenty million dollar deal that is about 1.67 per cent a year. On a hundred million dollar deal it is about 0.33 per cent. If the structure saves two points a year, the small deal keeps a third of a point, and the large one keeps 1.67 points. Size decides.

On accounting, under IFRS 9, the international standard for financial instruments, an originator derecognises receivables only if it transfers substantially all the risks and rewards, or gives up control. An originator that keeps the first loss usually keeps most of the risk. The receivables stay on its balance sheet and the notes become a financing liability. Assume off balance sheet treatment is wrong until the auditors confirm otherwise.

## Scene 7. What the model shows

*On screen: Inputs: structure 1 warehouse line, structure 2 securitisation on balance sheet. Investor IRR: default 46.4 to 46.5 per cent. SolaraPay 29.0 to 29.1 per cent.*

The companion workbook offers two facility structures on the Inputs sheet. Structure 1 is a warehouse line. Structure 2 is a securitisation or term asset backed issue, modelled on balance sheet, with its own rate, a haircut to advance rates and an upfront fee. True sale through an SPV is not modelled in version 0.8.

On that basis the option barely moves returns. For the default fictional company, the investor IRR, the internal rate of return, goes from 46.4 to 46.5 per cent. For SolaraPay, it goes from 29.0 to 29.1 per cent, with the multiple unchanged at 3.6 times and no breach months in either case. A structure cannot fix a pool. At most it prices a good one more efficiently, and SolaraPay's effort is better spent on collections and data.

## Scene 8. Recap and exercise

*On screen: Readiness: full tenor vintages, reconciled tape, stable definitions, legal opinion, backup servicer, sufficient size, tested triggers, auditor view. Exercise: T06 Loan tape.*

A securitisation moves receivables into an SPV and protects noteholders through subordination, excess spread and triggers, but in PAYGo it still depends on the originator as servicer. Excess spread is thin, fixed costs set a minimum size, and the receivables will probably stay on the balance sheet. In the model the structure changes returns by a

tenth of a point, so readiness matters more than structure.

Open the T06 loan tape template and load a sample of your own company's accounts, or of a company you are assessing. Run the validation sheet until every check count reads zero. Then list which of the eight readiness criteria in Chapter 12 your data can already support, and which would block an arranger mandate today.

## Module 13. RBF, subsidies and affordability

Running time: 10:01. Book: Chapter 13. Model sheets: Inputs (RBF block), RBF\_Engine, Vintage\_Input, Consumer\_Risk, Unit\_Economics, Sensitivity, Checks. Templates and tools: D1 Price plan and APR calculator, D2 Unit economics calculator.

### Learning objectives

1. Distinguish the four results a PAYGo subsidy can pay for and the incentive each one creates.
2. Use the four RBF designs in the workbook and read what the default model says about RBF value.
3. Measure an affordability gap and compare the ways of closing it, treating payment burden as a credit variable.

### Scene 1. Two decisions, one question

On screen: *Title card. Programme manager and investor, side by side. "How should subsidies be used when the product is credit?"*

This module serves two people. The first is a programme manager with public or philanthropic money, who must decide what to pay for, how to verify it, and how to stop the payment rewarding the wrong behaviour. The second is an investor or lender, who must decide how much weight to put on subsidy income when sizing returns and debt capacity.

Both decisions come down to how subsidies should be used in a business whose real product is consumer credit.

A subsidy to a PAYGo, or pay as you go, company does not buy a grid asset. It buys the start of a two to four year relationship with a household on irregular income. The benefit arrives only if the household keeps paying, or finishes paying and owns the device. That gap between what is paid for and what is wanted is the central design problem.

### Scene 2. What is being bought

On screen: *Four steps moving away from the sale. Sale, connection, repayment, ownership.*

Under results based financing, or RBF, the funder pays a fixed amount per verified result after the event. The company pre finances the activity and carries the execution risk. So everything depends on how the result is defined.

There are four candidates, each further from the sale. A sale is easy to verify, but says nothing about whether the customer will pay. A connection requires the system to be installed and working. SEforALL's Universal Energy Facility is described as paying per verified connection, though that is secondary reporting and its own documents must be read.

Repayment is the first result that carries credit information. Ownership is the closest to the public interest, and the slowest to observe. The PAYGo PERFORM 2026 standard defines it as the Ownership Rate at twice the contract term. On a 24 month contract, measured at twice the tenor, the first reading arrives four years after the first sale.

### Scene 3. Four designs in the workbook

On screen: *RBF\_Engine sheet. Mode 1 sales based, mode 2 repayment linked, mode 3 ownership linked, mode 4 hybrid.*

The RBF\_Engine sheet of the companion workbook holds four designs. RBF is set in dollars and recognised below gross profit when the cash arrives.

Take an illustration of twenty five dollars per verified Tier 2 unit, at one hundred and thirty local currency units to the dollar. Under mode 1 the company receives twenty five dollars, or three thousand two hundred and fifty local currency units.

Under mode 2, suppose the target is 80 per cent repaid at verification and the cohort has repaid 68 per cent. The payout factor is 68 divided by 80, which is 85 per cent. The company receives twenty one dollars and twenty five cents. Had the cohort repaid 84 per cent, the factor would be capped at 100 per cent.

Under mode 3, suppose forty dollars per owner at month 48, with 55 per cent owning. That is twenty two dollars per unit sold. Discounted at 20 per cent a year over four years, it is worth only about ten dollars and sixty one cents at the date of sale. A hybrid weighted 50, 30 and 20 per cent pays about twenty three dollars and twenty eight cents, most of it early.

#### Scene 4. Sales based RBF can reward bad credit

On screen: *Two bars. Company A, 1,000 units, 80 per cent repaid. Company B, 1,300 units, 60 per cent repaid.*

A sales based payment subsidises customer acquisition. The trouble is that the cheapest way to raise verified sales is to relax underwriting.

Consider two companies in the same programme. Company A underwrites tightly, sells 1,000 units and its cohort repays 80 per cent by verification. Company B underwrites loosely, sells 1,300 units and repays 60 per cent. Under sales based RBF, A receives twenty five thousand dollars and B receives thirty two thousand five hundred. The programme pays 30 per cent more to the company whose customers fall behind more often.

Under repayment linked RBF with an 80 per cent target, A still receives twenty five thousand dollars. B receives twenty four thousand three hundred and seventy five. The ranking reverses, but only modestly.

#### Scene 5. Repayment, ownership and hybrids

On screen: *Three short panels. Repayment: definitions. Ownership: timing and evidence. Hybrid: weights total 100 per cent.*

Repayment linked RBF lines the programme up with the lender. Both now want cohorts that pay. The data already sit on the lockout platform, but the definitions must be tight. They must say whether deposits count, whether written off accounts stay in the denominator, and whether the rate is measured by cohort at a fixed age or on the portfolio at a calendar date. The PAYGo PERFORM 2026 standard is the natural reference. Its repayment rate excludes deposits and subsidies, keeps written off contracts in, and is cumulative from contract start, computed on contract data. A collection rate is not a substitute. Because subsidies are excluded, RBF can never improve a company's reported repayment rate.

Ownership linked RBF has two weaknesses. The company waits years for the cash, and ownership is hard to prove, because an unlock can follow a settlement or a goodwill gesture. The workbook is strict. Mode 3 pays zero until validated ownership data sit in Vintage\_Input and the evidence switch is set to 1.

A hybrid pays some cash early and holds some back. The weights are a policy choice. The Checks sheet only confirms they total 100 per cent.

## Scene 6. What the model says RBF is worth

On screen: *Sensitivity sheet. Base investor IRR 46.4 per cent. RBF off 44.7 per cent. Repayment linked 46.4 per cent. Zero breach months.*

At default inputs, switching RBF off lowers the investor internal rate of return, or IRR, from 46.4 per cent to 44.7 per cent. That is a fall of 1.7 percentage points, with no covenant breach months either way. RBF helps, but it does not decide the case. A case that needs RBF to clear the hurdle is a bet on renewal.

Switching from sales based to repayment linked RBF leaves the IRR unchanged at 46.4 per cent. At Base credit quality the payout factor sits at or near its cap, so the change costs a good performer nothing. The design bites only where credit is weak.

## Scene 7. Paying the supplier or the customer

On screen: *Supplier subsidy versus consumer subsidy. Demand side voucher: eligibility, duplication, reconciliation.*

RBF paid to the company is a supplier subsidy. A twenty five dollar payment does not guarantee a twenty five dollar lower price, because the company may use it for acquisition or margin.

A consumer subsidy shows up in the price. It lowers the cash price or the instalment, and so lowers the payment burden directly. A demand side voucher routes that subsidy through the household. It keeps suppliers competing and targets more precisely, at an administrative cost. It must also fit the credit contract. A voucher that covers the deposit removes the one signal of willingness to pay. One that covers instalments makes the programme a co lender, and its terms must say what happens on default.

## Scene 8. Measuring an affordability gap

On screen: *Consumer\_Risk sheet. Income 30,000. Instalment 3,900. Burden 13.0 per cent. Threshold 10 per cent. Gap 900 a month.*

The Consumer\_Risk sheet compares each tier's instalment with an illustrative household income and flags any tier above a threshold, 10 per cent by default.

Take an illustration. A household earns thirty thousand local currency units a month. A Tier 2 instalment is three thousand nine hundred a month over 24 months, a burden of 13.0 per cent. To reach 10 per cent the instalment must fall to three thousand, a cut of nine hundred a month, or 23.1 per cent.

Extending the tenor lets default hazard act on more months. Raising the deposit excludes the households with least cash. You can also redesign the product, or buy the price down. At 50 per cent a year, nine hundred a month for 24 months has a present value of about thirteen thousand four hundred and ninety local currency units, well below the nominal twenty one thousand six hundred. Compare subsidy cost per completed contract, not per sale. And treat payment burden as a credit variable. It is available at origination, long before PAR30, the share of the book more than 30 days past due, starts to move.

## Scene 9. Recap and exercise

On screen: *Recap. Define the result. Model with and without RBF. Ownership needs evidence. Burden predicts loss. Exercise: Downside, mode 1 against mode 2.*

A subsidy is only as good as the result it pays for, and sales based payments can reward loose underwriting. Repayment linked RBF costs good performers nothing in the default model and redirects money towards credit discipline. Ownership income needs validated data before it appears in any case. Payment burden is a leading indicator of credit loss and belongs in the credit analysis.

Your exercise uses the workbook. On a copy, switch the scenario on Inputs to Downside, then run it once with RBF mode 1 and once with mode 2. Read the RBF line on RBF\_Engine and the investor IRR on Investment\_Summary for each. The difference between the two payouts is your measure of how much the Downside weakens credit quality.

## Module 14. Stress testing

Running time: 9:43. Book: Chapter 14. Model sheets: Inputs, Scenarios, Sensitivity, Covenants, Credit\_Portfolio, Valuation, Investment\_Summary, Dashboard. Templates and tools: D4 Receivables financing calculator, D6 Investor returns calculator, T05 Lender KPI report.

### Learning objectives

1. Design a Downside that moves credit, currency, volume, hardware cost and funding together.
2. Read the default scenario and sensitivity results, and judge which levers destroy equity and which hurt the lender.
3. Run a reverse stress on a covenant and on exit equity, and a funding drought, and read MOIC, peak equity and breach months together.

### Scene 1. What can kill the company

On screen: *Title card. "Base case: what management hopes. Stress test: how far away failure is."*

This module asks a blunt question, which is what can kill the company. A base case tells a committee what management hopes will happen. A stress test tells it which plausible combination of events would leave the company unable to pay its lenders, unable to raise its next round, or worth nothing at exit.

The committee does not need to believe the stress will happen. It needs to know how far away it is, and what can be done before the company gets there.

### Scene 2. How PAYGo companies fail

On screen: *Chain diagram. Collections fall, borrowing base shrinks, covenants approach limits, facility at risk.*

A PAYGo, or pay as you go, company usually runs out of cash while its income statement still looks acceptable. Hardware revenue is booked on the day of sale. The cash arrives over 12 to 48 months.

The book is funded partly by borrowing against eligible receivables. So when credit weakens, three things happen at once. Collections fall. The borrowing base shrinks, because more of the book ages beyond the eligibility cut off. And covenants on collection rate, receivables at risk and days past due, or DPD, move towards their limits. The lender's protection is triggered at the moment the company most needs the facility.

That is why stress testing here is mostly about credit and liquidity, with currency as the main amplifier. Volume, which worries founders most, is usually the least dangerous of the major variables.

### Scene 3. Designing a stress

On screen: *Scenarios sheet. Levers for Base, Downside, Severe: hazard 1.00, 1.30, 1.75. Collection 1.00, 0.97, 0.92. Volume 1.00, 0.90, 0.75. Hardware 1.00, 1.05, 1.10. Depreciation 5, 12, 25 per cent a year.*

The commonest mistake is to move one variable and call it a Downside. Real stress arrives in combinations. Depreciation raises the local cost of dollar priced hardware and often comes with inflation that squeezes household incomes. Squeezed households pay less reliably, and lenders grow cautious.

So cover five channels: credit, currency, volume, hardware cost and funding. The Scenarios sheet scales the first four. The Downside raises default hazard by 30 per cent, adds seven points of depreciation a year and trims volume by 10 per cent. The Severe case raises hazard by 75 per cent and has the currency losing a fifth of its value each year.

Two things are missing from the levers. The scenarios keep the facility in place, and they keep the default pricing response of a 5 per cent annual price increase and 50 per cent pass through of exchange rate moves. Test both separately.

#### Scene 4. The three scenarios at default inputs

On screen: *Table. Base: peak equity 10.0 million dollars, EBITDA margin 21.6 per cent, IRR 46.4 per cent, multiple 6.7 times, 0 breach months. Downside: 10.0, 7.3 per cent, minus 22.9 per cent, 0.3 times, 44. Severe: 46.6, minus 18.4 per cent, total loss, 0.0 times, 48.*

At default inputs, with an investor putting in four million dollars, the Base gives an investor internal rate of return, or IRR, of 46.4 per cent and a multiple of 6.7 times, with no covenant breaches.

The Downside is the instructive row. Peak equity does not move from ten million dollars. Yet the IRR is minus 22.9 per cent, the multiple is 0.3 times, so the investor loses about 70 per cent of its money. And the company is in breach of at least one covenant for 44 of 60 months. A cash survival test alone would miss both facts.

The Severe case adds a funding crisis. Peak equity rises to forty six point six million dollars, because the workbook tops up equity automatically to fill the gap. No shareholder group would supply thirty six point six million dollars more than Base. Read it as failure or a distressed sale.

#### Scene 5. One lever at a time

On screen: *Sensitivity sheet. Depreciation 20 per cent: minus 16.0 per cent. No price increase: 7.9 per cent. Hazard 1.5 times: 36.1 per cent, 40 breach months. Volume 20 per cent lower: 37.8 per cent, 0. Tier 4 and 5 mix: 66.6 per cent.*

The Sensitivity sheet moves one assumption at a time from Base. Five readings follow.

Currency and pricing destroy equity. Depreciation of 20 per cent a year takes the IRR to minus 16.0 per cent. Removing the 5 per cent annual price increase on new contracts takes it from 46.4 to 7.9 per cent. The price increase is how the company passes on dollar cost inflation.

Credit hurts the lender more than the equity. A 50 per cent higher default hazard leaves the investor at 36.1 per cent, but puts the company in breach for 40 months. Volume is the least dangerous lever. A 20 per cent fall gives 37.8 per cent and no breaches.

Structure barely moves returns. Securitisation adds 0.1 points. And the higher Tier 4 and 5 mix, at 66.6 per cent, is the most dangerous number in the table because it is the most attractive. It assumes larger systems repay at default parameters, without the cohort history to prove it.

#### Scene 6. Reverse stress testing

On screen: *Two calculations. Collection: 73.3 to 70 per cent, a fall of about 4.5 per cent. Equity: exit at 6.0 times EBITDA, net debt 4.0 billion, threshold EBITDA 0.667 billion.*

A forward stress asks what happens under a scenario. A reverse stress asks what scenario causes a given failure. It turns a scenario into a distance.

The default covenant needs a trailing three month collection rate of at least 70 per cent. In Base the annual rate falls from 84.5 per cent in Year 1 to 73.3 per cent in Year 5. A first pass says a uniform fall of about 4.5 per cent brings Year 5 to the limit. Then the workbook corrects you. With collections cut by 5 per cent, the company is in breach for 27 months, longer than the annual rates suggest. Locate the threshold by arithmetic, then confirm it in the workbook.

For equity, take an illustration. Suppose the company exits at 6.0 times EBITDA, its earnings before interest, tax, depreciation and amortisation, with Year 5 EBITDA of one billion local currency units and net debt of four billion. Equity is two billion. It is wiped out when EBITDA falls to about six hundred and sixty seven million, a fall of a third. If net debt also rises to four point five billion, the threshold rises to seven hundred and fifty million, a fall of only a quarter.

## Scene 7. Funding drought

*On screen: Inputs sheet, facility limit set to zero on a copy. Eligible receivables 4.0 billion times 70 per cent advance rate equals 2.8 billion, about 21.5 million dollars.*

A funding drought tests what happens when the facility is not there. GOGLA is reported to put off grid solar investment at about two hundred and ninety nine million dollars in 2024, down about 30 per cent. That is secondary reporting, still to be checked against the primary document.

To run it, set the facility limit to zero or its start month beyond the horizon, on a copy. Peak equity then measures the cost. As an illustration, eligible receivables of four billion local currency units at a 70 per cent advance rate give a facility of two point eight billion, about twenty one point five million dollars at one hundred and thirty to the dollar.

Failures do happen. BBOX LTD entered administration on 19 May 2025, and its latest filed accounts were for FY2022, according to the public register, with primary documents not yet reviewed. Filed accounts can be well over a year old by the time a failure is public.

## Scene 8. Reading total loss, and the SolaraPay stresses

*On screen: Three figures to read together: multiple, peak equity, breach months. SolaraPay: calibrated Downside total loss, 40 breach months. Full FX indexation: 23.6 per cent.*

IRR is a poor statistic in the tails. When exit equity is zero it cannot be computed, and the workbook reports a total loss. So read three figures together. The multiple says how much money comes back. Peak equity says how much had to go in. Breach months say how long the lenders were in control.

In the fictional SolaraPay case, the calibrated Downside is a total loss with 40 breach months, without any call for new cash. Full indexation of new prices to the exchange rate turns it into a 23.6 per cent IRR, though 38 breach months remain. Doubling the hazard on the untested Tiers 4 and 5 roughly halves the IRR, from 29.0 to 14.8 per cent.

## Scene 9. Recap and exercise

*On screen: Recap. Move levers together. Pricing and currency destroy equity. Credit hurts the lender. Measure the distance. Exercise: reverse stress on the collection covenant.*

A credible Downside moves credit, currency, volume and hardware cost together, and funding is tested on its own. In the default model, pricing and currency move returns more than any credit or volume assumption, while credit hits the lender hardest. Reverse stress turns scenarios into distances, and the multiple, peak equity and breach months must be read side by side.

Your exercise is a reverse stress. On a copy of the workbook, reduce the collection rate multiplier on Scenarios in small steps from 1.00, one change at a time. Record the multiplier at which the Covenants sheet first shows a breach month, and compare it with the first pass estimate of about 0.955. Then set the facility limit to zero and note the peak equity on Investment\_Summary.

# Module 15. The investor view: due diligence and the investment memo

Running time: 11:32. Book: Chapter 15. Model sheets: Credit\_Input, Vintage\_Input, Credit\_Portfolio, Vintage\_Dashboard, Valuation, Investment\_Summary, Investment\_Readiness, Covenants, Benchmark\_Compare, Calibration, Source\_Register. Templates and tools: T01 Investment committee memorandum, T02 Investor due diligence checklist, T06 Loan tape specification and data request, D5 Investment screening scorecard, D6 Investor returns calculator.

## Learning objectives

1. Organise a PAYGo diligence in eight workstreams, starting with credit and the data tape, and recognise the red flags that should pause a deal.
2. Value a business whose free cash flow is negative by design, and express returns in dollars.
3. Turn readiness gates and findings into a memo, conditions precedent and covenants.

## Scene 1. One asset, two contracts

On screen: *Title card. One receivables book. Equity: residual claim. Lender: senior claim.*

Every earlier module ends at the same decision: whether to invest or lend, and on what terms.

An equity investor in a PAYGo, or pay as you go, company buys a residual claim on a consumer credit book, a distribution network and a hardware margin. A receivables lender buys a senior claim on the same book, protected by an advance rate, eligibility criteria and covenants. The investor asks whether growth will compound into exit value. The lender asks whether, if growth stops tomorrow, the book will repay the facility.

Each needs the other's view. A covenant breach hands control of cash to the bank, and book equity is only as good as the loss allowance.

## Scene 2. Credit and the data tape come first

On screen: *Eight workstreams. Credit and data tape highlighted. Commercial, operations, technology, finance, legal, ESG, management.*

A full diligence has eight workstreams, and the overlaps are where problems hide.

Credit and the data tape is the core, and it should start first. The team takes an account level loan tape and the monthly portfolio and cohort data. It rebuilds the collection rate, receivables at risk, cohort curves and PAR30, the share of the book more than 30 days past due, and reconciles them to the ledger. The aim is to replace assumed behaviour with observed behaviour. In the SolaraPay case, a fictional company with synthetic history, recalibration on 24 months of history raised the Tier 1 to 3 default hazards by 30 per cent and cut the investor return from 33.1 to 29.0 per cent.

The commercial workstream produces the volume path and product mix. Scrutinise any shift towards larger systems, because mix moves value sharply on little evidence.

### Scene 3. Operations, platform and accounts

On screen: *Agents: clawbacks, concentration, fraud. Platform: unlock tokens and audit trail. Accounts: IFRS 9 staging at 30 and 90 days past due.*

Agents originate the credit. So diligence covers clawbacks on early default, concentration by agent and region, and controls against ghost sales and recycled devices. It also checks that repossession actually happens, because modelled recoveries need a real process.

The lockout platform is the collateral. Test who can issue unlock tokens, with what audit trail, and reconcile unlocks to payments. An unlock without full payment is a settlement, a goodwill gesture or a control failure.

Finance and accounting tests revenue recognition and the loss allowance. Under IFRS 9, the international accounting standard for financial instruments, check the staging. Stage 2 follows a significant increase in credit risk, presumed at 30 days past due. Stage 3 follows default, presumed at 90. An allowance that stays flat while PAR30 rises is a finding in its own right.

Legal work maps licences against the rules, confirmed with local counsel. Environmental, social and governance work covers payment burden and the conduct of collections. Management diligence asks whether the credit function can slow sales.

### Scene 4. The data request and the red flags

On screen: *Template T06 data request. Red flags: buckets that do not reconcile, deposits in the collection rate, re ageing, unexplained unlocks, flat allowance, high recoveries, concentration, cases that need RBF.*

Issue the data request on day one. The minimum is the account level tape, 24 months of portfolio data by tier in the structure of Credit\_Input, cohort observations in the structure of Vintage\_Input, ledger reconciliations, unlock logs and every financing and subsidy agreement. Template T06 sets this out.

Some findings should pause a deal until explained. Days past due buckets that do not add up to gross receivables. A collection rate that includes deposits without saying so. Restructurings that reset days past due, especially just before a covenant date. Unlocks without matching payment. A loss allowance that does not move with PAR30. And an investment case that clears the hurdle only with subsidy income, a richer mix of untested tiers or a generous exit multiple.

One flag outweighs the rest. A team that cannot produce its cohort data within two weeks is unlikely to be managing to it.

### Scene 5. Valuing a business with negative free cash flow

On screen: *Terminal value formula on normalised cash flow. Illustration: 1.1 times 1.05, less 0.05 times 6.0, divided by 0.17, equals about 5.03 billion.*

Free cash flow is EBITDA, meaning earnings before interest, tax, depreciation and amortisation, less tax, capex and the increase in working capital. Here working capital is mostly the receivables book, so while it grows, free cash flow is negative and value sits in the terminal year.

The workbook reinvests only what the book needs to grow at the long run rate. Take an illustration. Year 5 EBITDA is one point five billion local currency units, less tax of three hundred million and capex of one hundred million, leaving one point one billion. Working capital is six billion, growth is 5 per cent and the discount rate is 22 per cent. One point one times 1.05 is 1.155. Take away 5 per cent of six, which is 0.3, and you have 0.855. Divided by 0.17, that gives a terminal value of about five point zero three billion. On raw Year 5 free cash flow it would have been negative.

Even normalised, the discounted cash flow, or DCF, understates a growth book. In the SolaraPay case it gives six point one million dollars at 22 per cent, below the eight million dollar pre money valuation.

### Scene 6. Exit, book value and dollar returns

*On screen: Valuation sheet. Exit at 4.0 times EBITDA: IRR 46.4 falls to 33.9 per cent. Depreciation over five years: 5 per cent leaves 78.4 per cent, 12 per cent leaves 56.7 per cent, 25 per cent leaves 32.8 per cent.*

The exit valuation applies a multiple of EBITDA and deducts net debt. In the default model, exiting at 4.0 times EBITDA instead of the base multiple lowers the investor internal rate of return, or IRR, from 46.4 to 33.9 per cent.

Price to book treats the company as a lender, so it is only as reliable as the loss allowance.

Dollar investors are judged in dollars. For SolaraPay, four million dollars at eight million pre money buys 33.3 per cent. A third of exit equity of forty two point nine million dollars is fourteen point three million, a multiple of about 3.6 times and an IRR of about 29.0 per cent. Currency deserves its own line in the memo. Over five years, depreciation of 5 per cent a year leaves 78.4 per cent of the value, 12 per cent leaves 56.7 per cent, and 25 per cent leaves 32.8 per cent.

### Scene 7. Gates, the memo and conditions

*On screen: Investment\_Readiness banner: "3/23 gates met. Decision: STOP (Failed test: No covenant breach in the active scenario)". Template T01, twelve sections, recommendation first.*

The Investment\_Readiness sheet lists 23 gates, some automatic and some set on external evidence. A manual gate counts only when the sheet records where the evidence is held and who signed it off. Thirteen gates are critical, and the sheet applies a fixed rule. STOP when a test fails: the master check, a covenant breach in any month or a year with the debt service coverage ratio below its minimum, or a tier with negative contribution. GO only when all 23 gates are met. CONDITIONAL GO when every critical gate is met, with the open gates as conditions. Otherwise STOP, because the evidence is incomplete. A STOP on incomplete evidence is not a verdict on the business, but a STOP on a failed test is a finding against the plan as modelled. A GO is not an investment recommendation. At default inputs two gates are met and the rule reads STOP on a failed test: no month breaches a covenant, but the annual DSCR sits below 1.20 times in Years 1 to 4. In the SolaraPay case four gates are met, and the rule reads STOP for the same reason.

The memo in template T01 follows twelve sections, with the recommendation and its conditions first. Show the DCF and exit values side by side.

Write changes as instruments, not hopes. Conditions precedent fix what must be true before money moves, such as a reconciled tape or an independent review of expected credit loss, or ECL. Covenants fix what must stay true. Prefer portfolio covenants to a debt service coverage test during growth, set them at levels the history supports, and add a cure period. An unmet gate is often a condition to attach.

## Scene 8. Benchmarks and their limits

On screen: *Benchmark\_Compare, Calibration and Source\_Register. Grades A to D, status recorded separately. M-KOPA comparison withdrawn until the group's consolidated accounts are read.*

Benchmarks are a sanity check, not a target. The workbook grades each source from A, for a primary official or audited document, to D, for unverified material, and records separately whether the claim has been verified. Calibration uses a reference only once its source is verified.

M-KOPA shows why that discipline matters. Two outlets reported FY2024 group revenue figures that differ by a factor of about 1.64. The only filing held is that of M-KOPA UK LIMITED, a subsidiary, which says nothing about the group's PAYGo business. So the comparison is withdrawn until the group's consolidated accounts are read, and neither revenue figure is used as a benchmark. The ESMAP Off-Grid Solar Market Trends Report 2024 is reported to give an average PAYGo collection rate of about 62 per cent for 2021 to 2023. It is pending the primary document, and it is a collection rate, not a repayment rate. The benchmark that matters most is the company's own cohort history.

## Scene 9. Recap and exercise

On screen: *Recap. Credit first. Red flags pause the deal. Value on normalised cash flow and in dollars. Gates become conditions. Exercise: D6 and T01.*

Start with credit and the data tape, and let the red flags stop a deal until they are explained. Value a growing book on normalised cash flow, read the DCF and exit values together, and translate returns into dollars. Turn every unmet readiness gate into a condition precedent, a covenant or an accepted risk.

Your exercise uses the D6 Investor returns calculator, which opens on the SolaraPay calibrated Base at 3.57 times and 29.0 per cent. Raise the depreciation rate from the Base path to the Downside rate of 12 per cent a year and record the new IRR. Then use the solver to find the highest pre money valuation that meets your fund's hurdle rate. Write both results into section 8 of the T01 memo, with one sentence on what the entry price assumes.

## Module 16. Case walk through: SolaraPay

Running time: 12:36. Book: Chapter 16. Model sheets: Credit\_Input, Vintage\_Input, Credit\_Assumptions, Credit\_Portfolio, Vintage\_Dashboard, Products, Unit\_Economics, Consumer\_Risk, Dashboard, KPIs, Covenants, Valuation, Benchmark\_Compare, Calibration, Investment\_Readiness. Templates and tools: case workbook SolaraPay\_Case\_Model\_v0.8-dev, D3 Repayment curve and cohort calibration, D6 Investor returns calculator, T01 Investment committee memorandum, T06 Loan tape specification and data request, T07 Borrowing base certificate and term sheet checklist.

### Learning objectives

1. Apply the method end to end: load history before touching projections, recalibrate, and read unit economics and affordability.
2. Compare the management plan with the calibrated case under stress, and read valuation and the lender's covenant position.
3. Build a Conditional Go recommendation whose conditions map to the risks and readiness gates found.

### Scene 1. The ask

On screen: *Title card. SolaraPay Ltd, Republic of Kivara, fictional. Equity: 4.0 million dollars at 8.0 million pre money, 33.3 per cent. Facility: 4.5 billion shillings at 16 per cent.*

This module follows an analyst at an impact fund through one file. SolaraPay Ltd and the Republic of Kivara are fictional, the 24 month history is synthetic, and every figure comes from the case workbook.

SolaraPay sells solar products on PAYGo, or pay as you go, terms. Its currency is the Kivara shilling, at an opening rate of one hundred and thirty five shillings to the dollar. It asks the fund for four million dollars of equity at a pre money valuation of eight million, a stake of 33.3 per cent. It asks a local bank for a receivables facility of four point five billion shillings at 16 per cent. Eligibility runs to 30 days past due.

About 65 per cent of planned volume sits in Tiers 1 and 2. Most of the value sits in Tiers 4 and 5, which the company has never sold. And the facility is about 3.7 times total equity in shillings, so the equity case depends on the facility and its covenants.

### Scene 2. Load the history first

On screen: *Credit\_Input and Vintage\_Input loaded. Credit data mode set to Actual on Credit\_Assumptions. Portfolio collection rate 77.4 to 69.3 per cent. Latest month: 66.5 per cent, PAR30 17.0 per cent.*

Management built its plan on the workbook's default credit assumptions. The analyst loads the company's data before touching the projections.

Twenty four months of portfolio data for Tiers 1 to 3 go into Credit\_Input, and eighteen monthly cohorts per tier into Vintage\_Input. The credit data mode is set to Actual. That changes the reports, not the projections. History is never blended silently into the forecast.

The data contradict the plan. The portfolio collection rate fell from 77.4 per cent in the first year to 69.3 per cent in the second. In the latest month the collection ratio is 66.5 per cent, PAR30, the share more than 30 days past due, is 17.0 per cent, and indicative coverage by expected credit loss, or ECL, is 36.5 per cent. At month 12, cohorts repay

below plan in every tier. Tier 2, for example, has repaid 69.4 per cent against a plan of 74.5. Recoveries are close to nil, with a loss given default proxy of about 99 per cent. And no cohort is old enough to show ownership.

### Scene 3. Recalibration

On screen: *Products sheet. Hazard: Tier 1 3.50 to 4.55 per cent, Tier 2 2.60 to 3.38 per cent, Tier 3 1.90 to 2.47 per cent. Survival to end of contract, Tier 2: 53.1 to 43.8 per cent.*

The analyst re-estimates the Tier 1 to 3 curves so that projected cohorts sit on the observed points. Each calibrated monthly default hazard is exactly 1.30 times the plan. That is the workbook's Downside multiplier. On credit hazard, the company's history already sits at management's Downside. Collection rates on paying accounts fall by one or two points.

The effect is large. For Tier 2 over 24 months, the share of accounts still paying at the end falls from 53.1 per cent to 43.8 per cent. Tiers 1 and 3 lose eight to nine points.

Tiers 4 and 5 stay on the default curves, because there is no history. That assumes customers of products never sold will repay better than any tier the company has sold. Real data will fit less neatly than synthetic history, but the principle holds: a plan contradicted by the company's own history cannot serve as a base case.

### Scene 4. Unit economics and affordability

On screen: *Unit\_Economics table. Expected loss, LTV to CAC, payback, unit IRR by tier. Tier 2: 41.8 per cent, 3.5 times, 17 months, 37 per cent. Consumer\_Risk: Tier 2 13.0, Tier 3 13.2, Tier 4 11.3 per cent of income.*

With calibrated curves, every tier still creates value per unit. The distribution is the issue.

Tier 2 is the largest product, at 40 per cent of planned units, and the weakest. It loses 41.8 per cent of scheduled instalments, pays back in 17 months, and earns a unit internal rate of return, or IRR, of 37 per cent, the lowest in the range.

Tiers 4 and 5 look strongest, with lifetime value to acquisition cost, or LTV to CAC, of 11.0 and 14.9 times. Those ratios rest entirely on curves with no history. Tier 1 pays back in nine months, but its contribution is only one thousand and ninety two shillings per unit, and its implied annual rate of 106 per cent will draw attention.

Affordability adds a second lens. Three tiers sit above the model's 10 per cent policy threshold for payment burden: Tier 2 at 13.0 per cent of illustrative income, Tier 3 at 13.2 and Tier 4 at 11.3. The incomes are illustrative and need survey data.

### Scene 5. The calibrated projections

On screen: *Dashboard. Revenue 1.13 to 8.62 billion shillings. EBITDA margin minus 35.2 to 16.4 per cent. Collection rate 82.3 to 70.3 per cent. Debt to book equity peaks at 2.51 times in Year 3.*

In the calibrated Base, revenue grows from one point one three billion shillings in Year 1 to eight point six two billion in Year 5, or from eight point one to fifty one point two million dollars.

The margin on EBITDA, earnings before interest, tax, depreciation and amortisation, moves from minus 35.2 per cent to 16.4 per cent. Net credit losses still take 27.5 per cent of revenue in Year 5. The collection rate falls every year and ends at 70.3 per cent, a fraction above the proposed covenant. Leverage peaks at 2.51 times book equity

in Year 3, inside the limit of 3.0 times.

No equity top up is needed. Against the management plan, calibration lowers the investor IRR from 33.1 to 29.0 per cent. The plan survives its own data but loses most of its headroom.

## Scene 6. Stress tests

*On screen: Stress table. Calibrated Downside: total loss, 40 breach months. Full FX indexation: 23.6 per cent, 38 months. Tier 4 and 5 hazard doubled: 14.8 per cent. Entry at 6 million pre money: 33.8 per cent.*

The calibrated Downside applies the 1.30 hazard multiplier on top of a hazard already 1.30 times the plan. Relative to management, Tier 1 to 3 hazards are 1.69 times the plan, close to the 1.75 of the Severe case. On this data, the Downside may be more likely than its label suggests.

The Downside wipes out the equity without any call for new cash. Peak equity stays at nine million dollars, but at exit net debt exceeds six times a much reduced EBITDA, so exit equity is zero, with 40 breach months.

Pricing power is the strongest mitigant. Indexing new contract prices fully to the exchange rate turns the Downside into a 23.6 per cent IRR and a 14.2 per cent EBITDA margin. It leaves 38 breach months, so it protects the equity more than the lender.

Tier 4 and 5 credit is the largest unproven risk. Doubling their hazard roughly halves the IRR, from 29.0 to 14.8 per cent. Entry at six million pre money lifts the IRR to 33.8 per cent, which helps but does not change the risk.

## Scene 7. Valuation and the lender view

*On screen: Valuation sheet. DCF enterprise value 6.1 million dollars at 22 per cent. Exit equity 42.9 million dollars. One third is 14.3 million, a multiple of 3.6 times. Lowest trailing collection rate 70.0 per cent against 70 per cent.*

The discounted cash flow, or DCF, gives an enterprise value of six point one million dollars at 22 per cent, below the eight million pre money. A third of exit equity of forty two point nine million dollars gives a multiple of about 3.6 times and an IRR of about 29.0 per cent. The entry price is a position on the exit multiple and on credit quality, and the fund should protect both.

For the bank, the monthly covenants hold in the calibrated Base, but barely. The lowest trailing three month collection rate is 70.0 per cent against a 70 per cent minimum, so there is no headroom. On actual data, with 69.3 per cent over the second year and 66.5 per cent in the latest month, a 70 per cent covenant would very likely be breached at the first test date. The bank can set the covenant at a level the book can meet, such as 65 per cent with a cure period, or defer the facility. The debt service coverage ratio sits below its 1.20 times minimum in Years 1 to 4, so that covenant fails as drafted, and portfolio covenants should replace it. Monthly reporting should cover the operational and lender metrics, each with its definition printed, and the PERFORM 2026 KPIs once the company computes them on its contract data.

## Scene 8. Benchmarks and the recommendation

*On screen: Benchmarks: no verified reference, M-KOPA comparison withdrawn. ECL to financing revenue 1.28 times. Analyst: Conditional Go, seven conditions. Workbook: 4 of 23 gates, STOP on a failed test (annual DSCR).*

The analyst finds that no external reference can be used as a test. The net margin and credit loss references rested on M-KOPA group figures that conflict between sources, so that comparison is withdrawn until the group's consolidated accounts are read. The ratios speak for themselves. Expected credit loss runs at 1.28 times financing revenue, so the financing income does not cover the losses it is meant to price. The hardware margin is carrying them. The benchmark that matters is SolaraPay's own cohort history.

The analyst recommends a Conditional Go: invest four million dollars subject to seven conditions. Index new prices fully to the exchange rate. Cap Tiers 4 and 5 at their planned 13 per cent of units, released only after twelve months of cohort data within 10 per cent of plan. Agree a collections plan and a 65 per cent trailing collection covenant with a cure period. Cut recovery assumptions to observed levels. Redesign Tier 2 to 4 price plans so the instalment stays within 10 per cent of surveyed income. Protect the valuation, towards six million pre money or with a ratchet. And require monthly data, ownership tracking, an independent ECL review and a full test of the workbook.

The workbook's own rule reads STOP for the same file. The case meets 4 of the 23 readiness gates, and the covenant gate fails because the annual DSCR sits below its minimum in Years 1 to 4. That STOP is a finding against the plan as modelled, not only missing evidence. The two still do not contradict each other. The workbook applies a fixed rule to the plan and the evidence; the recommendation is the analyst's judgement of what the investment needs. So the Conditional Go is conditional on replacing or resetting the DSCR covenant with the lender, and disbursement waits until the critical gates that the conditions address are evidenced. The committee should see both, and the memo says so.

## Scene 9. Recap and exercise

*On screen: Recap. History before projections. Calibrated Base sits at plan Downside. Pricing protects equity. Tiers 4 and 5 unproven. Exercise: combined stress, recorded in T01.*

Load history first, and treat the management plan as one scenario among several. On SolaraPay's own data the Base sits at the plan's Downside on credit hazard, and the collection covenant has no headroom. Pricing power protects the equity, while the untested Tiers 4 and 5 hold both the strongest unit economics on paper and the largest unproven risk. The most important number in a PAYGo file is often the one the company cannot yet produce.

Your exercise uses the case workbook. On a copy, run the calibrated Downside with full price indexation and, at the same time, double the default hazard on Tiers 4 and 5. Record the IRR, the multiple and breach months. Then decide whether the seven conditions still support a Conditional Go, and write your answer in section 6 of the T01 memo.

## Module 17. Using MODEL 2, step by step

Running time: 19:34. Book: Chapters 1 to 16. Model: MODEL 2, version 0.8 (development build), file AEF\_SHS\_PAYGo\_Model\_v0.8-dev.xlsx. User manual: Steps 1 to 12.

### Learning objectives

1. Work through the model in the order an analyst should.
2. Know which sheet holds each input and which sheet answers each question.
3. Read the integrity checks, the readiness gates and the investment summary correctly.

### Scene 1. Using MODEL 2, step by step

On screen: *PAYGo Company Financial and Investment Model, in twelve steps*

This tutorial walks you through MODEL 2, the PAYGo Company Financial and Investment Model, in twelve steps, in the order an analyst should work. Every screen you will see is the real workbook, version 0.8, a development build, with its default inputs. Those inputs describe a fictional company in a fictional market. They are there to make the mechanics visible, not to describe any real business. Keep the user manual open beside you. It follows the same twelve steps.

### Scene 2. Start here: Cover

On screen: *Open the workbook, let it recalculate, and read the status panel before anything else. Sheet Cover, range B16:J44.*

Open the workbook in Excel 2016 or later and let it recalculate fully. The cover carries a live status panel. First, the master check. It must read OK before you rely on any output. Below it, the readiness line: two of twenty three gates are met at default inputs, and the decision reads STOP, on a failed test. The covenant gate fails, as Step 11 shows. The model never rates a company. Then the active case, the illustrative model assumptions in the Base scenario, and the credit data mode, Proxy, which means the credit figures come from the model's curves. Before you change anything, save a copy under a new name and keep the original as your reference.

### Scene 3. Start here: Start

On screen: *One page: what to input, what the model calculates, what the results mean and what an investor should look at. Sheet Start, range A1:D40.*

Next, the Start sheet. It sets out on one page what to input, what the model calculates, what the results mean, and what an investor should look at, in that order. Every input carries a provenance label: model assumption, company data, external evidence, calibrated assumption or unverified. Read the third block before you quote any number. Revenue is not cash. The operational collection rate is not a PAYGo PERFORM KPI. The credit loss figures are analytical proxies. And the readiness decision measures the evidence file, not the business.

## Scene 4. Start here: Checks

On screen: *Thirty eight integrity tests roll up into the master check. Readiness flags sit apart. Sheet Checks, range A1:C54.*

The Checks sheet runs thirty eight integrity tests. The balance sheet balances, cash reconciles every month, the DPD buckets reconcile to gross receivables, and receivables, the loss allowance, debt and equity roll forward from month to month. Limits are respected and inputs are valid. Each test returns zero when it passes, and together they drive the master check. Below them sit seven readiness flags, such as affordability assumptions not yet reviewed. They are kept outside the master check on purpose. They tell you how mature the analysis is, not whether the arithmetic is right.

## Scene 5. Start here: Inputs

On screen: *Blue cells are inputs, each with a provenance label. Set the scenario selector, the first model month and the currency label. Sheet Inputs, range A1:E14.*

All hard coded assumptions live on a handful of sheets, and the colour code tells you where. Blue font is an input, the only kind of cell you should change. Black font is a formula. Never overwrite it. On the Inputs sheet, start with three settings. The first is the scenario selector, where one is Base, two is Downside and three is Severe. Then come the first model month and the local currency label. The last column gives each input's provenance. At default inputs every one reads model assumption, because the defaults describe a fictional market.

## Scene 6. Define the business: Products

On screen: *Five tiers, labelled by the capacity attribute of the Multi Tier Framework. Replace them with your own catalogue. Sheet Products, range A4:G16.*

The Products sheet holds one column per tier. Tier one is a pico solar kit on a twelve month plan. Tier five is a three kilowatt peak inverter system on a forty eight month plan. Replace the names, sizes, loads and segments with your own catalogue. Keep all five columns. If you sell fewer tiers, set the sales mix of the unused tier to zero. The tier label refers to capacity only, so check it against the product's real specification.

## Scene 7. Define the business: Consumer\_Risk

On screen: *Affordability is a credit risk. Replace the illustrative incomes with surveyed incomes. Sheet Consumer\_Risk, range A1:G15.*

Consumer Risk sets each tier's instalment against the household income of its target segment. The incomes shown are illustrative placeholders and must be replaced with survey or customer data. With the defaults, three tiers sit above the model's ten per cent policy threshold for payment burden: Tier 2 at 13.0 per cent, Tier 3 at 13.2 per cent and Tier 4 at 11.3 per cent. A high burden predicts higher default, so treat these flags as credit warnings, not only as social ones.

## Scene 8. Market assumptions: Inputs

On screen: *FX, inflation, tax, the price increase on new contracts and FX pass through. Sheet Inputs, range A9:D14.*

The macro block sets the opening exchange rate, 130 local currency units per dollar, local inflation of six per cent, a tax rate of thirty per cent, an annual price increase of five per cent on new contracts, and a pass through of fifty per cent of currency depreciation into new contract prices. Do not set depreciation to zero because the currency has been stable. A PAYGo company buys hardware and borrows in dollars while it collects in local currency.

## Scene 9. Market assumptions: Scenarios

On screen: *Downside and Severe move credit, collections, volume, hardware cost and the currency together. Sheet Scenarios, range A4:E12.*

The Scenarios sheet scales five levers. The Downside raises the default hazard by thirty per cent, trims collections and volume, raises hardware cost and sets depreciation at twelve per cent a year. The Severe case goes further, with a hazard 1.75 times Base and twenty five per cent depreciation. Real stress moves several variables at once, and so do these scenarios.

## Scene 10. Build demand: Inputs

On screen: *Total units sold for Years 1 to 5 across all tiers. Sheet Inputs, range A16:D21.*

Enter the total units sold in each of the five years. The defaults run from twelve thousand units in Year 1 to fifty thousand in Year 5. The model divides each year by twelve, applies the scenario volume multiplier and splits the sales by tier. Version 0.8 does not model seasonality, so read monthly results as averages.

## Scene 11. Build demand: Products

On screen: *The sales mix must total 100 per cent. A check enforces it. Sheet Products, range A4:G23.*

The sales mix sits on Products, and it must total one hundred per cent. A check enforces it. Be careful when you shift the mix towards Tiers 4 and 5. It raises value sharply, but those tiers usually have the least repayment history behind them.

## Scene 12. Build revenue: Products

On screen: *Cash price, deposit, daily rate and tenor; the model derives the instalment, the premium and the implied APR. Sheet Products, range A12:G39.*

Each price plan has four inputs: cash price, deposit, daily rate and tenor. Take Tier 2. A cash price of 39,000, a deposit of 4,000 and a daily rate of 77 over 24 months give a monthly instalment of 2,342 and a total contract value of 60,210. The PAYGo premium over the cash price is 21,210, and the implied annual rate is about fifty per cent. That rate is the figure a regulator or a journalist will compute, so know it first. Hardware revenue is booked at the cash price at sale, and the premium is earned as financing income over the tenor.

### Scene 13. Build revenue: Inputs

On screen: *Four RBF designs. Ownership linked RBF pays nothing until validated ownership data exist. Sheet Inputs, range A30:D39.*

Results based financing has four designs: sales based, repayment linked, ownership linked and hybrid. Ownership linked RBF pays zero by design until validated ownership data have been loaded and this evidence switch is set to one. The model will not manufacture that outcome from proxy curves.

### Scene 14. Hardware and capital costs: Products

On screen: *Hardware FOB cost in US dollars; landed cost includes freight and duty at the month's exchange rate. Sheet Products, range A17:G46.*

Hardware is entered as the dollar FOB cost per unit. The model adds freight and duty, converts at the exchange rate of the month and applies the scenario hardware cost lever. For Tier 2, a cost of 150 dollars becomes a landed cost of 23,400 at the opening rate. Installation, warranty and the acquisition cost are set per tier. Do not enter a local price already converted at today's rate. That would count depreciation twice.

### Scene 15. Operating costs: Inputs

On screen: *Payment fees, servicing per active account, staff and general and administrative costs. Sheet Inputs, range A23:D28.*

Operating costs follow their drivers. Payment fees are two per cent of cash collected. Customer service and collections cost 150 per active account per month. Staff and general costs are fixed monthly amounts, indexed to inflation. Size the central costs for the volume plan. A book of seventy thousand active accounts needs a collections team, a call centre and a data function.

### Scene 16. Financing: Inputs

On screen: *Initial equity, a dollar term loan and a local currency receivables facility, with an automatic equity top up. Sheet Inputs, range A48:D57.*

Four sources fund the plan: initial equity, a dollar term loan revalued every month, a local currency receivables facility, and an automatic equity top up whenever cash would fall below the minimum. The facility is drawn only to hold the minimum cash balance, up to the lower of its limit and the borrowing base. The cumulative top up is your funding requirement. It is equity you must raise, not free money.

### Scene 17. Financing: Credit Assumptions

On screen: *Default at 180 DPD, staging thresholds, borrowing base eligibility, recovery cost and the proxy DPD distribution. Sheet Credit\_Assumptions, range A4:E24.*

Credit Assumptions holds the definitions. Default at 180 days past due. Stage 2 from thirty days and Stage 3 from ninety days for the indicative expected credit loss. Borrowing base eligibility up to thirty days past due. A recovery cost of fifteen per cent of resale proceeds. The binding constraint on the facility is usually the borrowing base, not the limit, and it shrinks when credit quality deteriorates.

### Scene 18. Financial statements: Annual

On screen: *Read the revenue mix, credit losses charged at origination, and EBITDA year by year. Sheet Annual, range A5:I28.*

The Annual sheet sums the monthly statements by year. Check the revenue mix between hardware, financing income and other revenue. Expected credit losses are charged when each contract is originated, so a growing book carries the full lifetime loss of its newest cohorts. In the default Base case, EBITDA turns positive in month 21.

### Scene 19. Financial statements: Annual

On screen: *Profit arrives before cash. The balance check must read zero; the cash flow shows the true funding gap. Sheet Annual, range A30:I62.*

Now the balance sheet and cash flow. The balance check must read zero in every period. Operating cash flow stays negative while the receivables book grows. In the default case it stays positive only from month 51, thirty months after EBITDA. A PAYGo company can report a profit and still run out of cash. The cash flow statement and the funding requirement are the figures to trust.

### Scene 20. Scenarios: Sensitivity

On screen: *Fifteen dated cases computed at default inputs, and a live row for the active case. Currency and pricing move value more than volume. Sheet Sensitivity, range A5:I24.*

Switch the selector between Base, Downside and Severe and read the investment summary each time. The Sensitivity sheet holds a static table of fifteen cases, dated and computed at default inputs, and below it a live row that follows the active case. The Base investor IRR is 46.4 per cent. In the Downside the IRR is minus 22.9 per cent, and the Severe case is a total loss. Look at the currency rows. Twenty per cent depreciation a year takes the Base IRR to minus sixteen per cent, and freezing prices on new contracts cuts it to 7.9 per cent. Remember to reset the selector to Base before you save.

### Scene 21. Lender case: Credit\_Portfolio

On screen: *Gross receivables by DPD bucket, the collection ratio, indicative ECL and the borrowing base with its headroom. Sheet Credit\_Portfolio, range A4:L31.*

Credit Portfolio consolidates the five tiers month by month: gross receivables by DPD bucket, the collection ratio, PD and LGD proxies, the indicative stage based ECL, and the borrowing base with its headroom against the facility. The buckets reconcile exactly to gross receivables. This is the view a lender reads first.

## Scene 22. Lender case: Covenants

On screen: *Each covenant is tested monthly. Headroom matters as much as compliance. Sheet Covenants, range A4:L26.*

Covenants tests each threshold every month and flags breaches. In the default Base case the lowest trailing three month collection rate is 72.9 per cent against a minimum of seventy, and no month breaches. Look at the headroom, not only at the pass or fail flag. The annual DSCR sits on the KPIs sheet. It is below its minimum of 1.20 times in Years 1 to 4, so the readiness gate on covenants is not met. DSCR is a poor test for a growing book, so replace it with portfolio covenants in the facility rather than ignore it.

## Scene 23. Lender case: Credit\_Input

On screen: *Paste the company's own history, then switch the credit data mode to Actual. Shown here: the fictional SolaraPay case. Sheet Credit\_Input, range A6:L25.*

Now load the company's own data. Credit Input takes one block per tier and up to sixty months: month end balances and DPD buckets, and monthly flows such as originations, collections and write offs. The screen shows the SolaraPay case workbook, whose history is synthetic teaching data. The buckets must add up to gross receivables. Then set the credit data mode to Actual. Reporting switches to the company's history, while projections stay on the curves.

## Scene 24. Lender case: Vintage\_Dashboard

On screen: *Compare observed cohorts with the proxy curves, then recalibrate the hazards and collection rates on Products. Sheet Vintage\_Dashboard, range A1:I20.*

Vintage Dashboard sets the observed cohorts against the plan curves. In the SolaraPay case, every tier with history repays below plan at month 12. When the curves diverge, recalibrate the default hazard and collection rates on the Products sheet. Lenders lend against data, and twelve or more months of clean cohort history is the fastest route to better terms.

## Scene 25. Investment memo: Valuation

On screen: *DCF on normalised free cash flow, exit value, and investor returns in US dollars. Sheet Valuation, range A22:D40.*

Valuation holds three views. The DCF of free cash flow gives an enterprise value of about 15.6 million dollars, with a terminal value larger than the whole value, which is normal for a growing PAYGo book. The exit value at six times EBITDA gives an equity value of about 80.6 million dollars at the end of Year 5. The investor's stake of one third then returns an IRR of 46.4 per cent and 6.7 times the money, in dollars. Present the DCF and the exit value side by side and explain the gap.

## Scene 26. Investment memo: Unit\_Economics

On screen: *Per tier: expected loss, contribution, LTV to CAC, cash payback and unit IRR. Sheet Unit\_Economics, range A4:G24.*

Unit Economics answers whether each sale creates value. For Tier 2 at default inputs, the expected loss is about 35.6 per cent of scheduled instalments, LTV to CAC is 4.8 times, and the cash payback is fifteen months. Read the ratio with the payback and the unit IRR. A strong ratio on a product with no repayment history is a hypothesis, not a result.

## Scene 27. Investment memo: Calibration

On screen: *Diagnostics use verified references only. Suspended or conflicting references are shown, never used as targets. Sheet Calibration, range A5:F12.*

Calibration turns benchmarks into questions, and it compares the model only with references whose source is verified. At this edition, every reference on the sheet is suspended or absent. The M-KOPA group figures conflict between sources, so that comparison is withdrawn until the group's consolidated accounts are read. The ESMAP sector collection rate is pending its primary document, and it is context for the model's operational collection rate, not a PERFORM repayment rate. The sheet asks you to justify the cost and credit assumptions on the company's own data. It never changes an input on its own.

## Scene 28. Investment memo: Investment\_Readiness

On screen: *Twenty three gates, thirteen critical. The decision rule reads evidence only: STOP, CONDITIONAL GO or GO. Sheet Investment\_Readiness, range A4:G36.*

Finally, Investment Readiness. Twenty three gates, automatic where the model can test them and manual where they need outside evidence, such as legal review or the auditor's view of the ECL approach. A manual gate counts only when the sheet records where the evidence is held and who signed it off. Thirteen gates are critical, and the decision rule uses the evidence and nothing else. It reads STOP when a test fails, GO only when all twenty three gates are met, and CONDITIONAL GO when every critical gate is met. Otherwise it reads STOP, because the evidence is incomplete. At default inputs two gates are met and the decision is STOP on a failed test. Gate ten now covers the annual DSCR as well as the monthly covenants, and the DSCR sits below its minimum in Years 1 to 4. A STOP on incomplete evidence is not a verdict on the business, but a STOP on a failed test is a finding against the plan as modelled. A GO is not an investment recommendation. Every gate not yet met should become a condition precedent, a covenant or an accepted risk in the investment memo.

## Scene 29. Investment memo: Investment\_Summary

On screen: *The one page summary for the investment committee, live for the active scenario. Sheet Investment\_Summary, range A4:G35.*

The Investment Summary brings it together on one page for the active scenario: the operating trajectory, the funding requirement, valuation and returns, the lender view and unit economics by tier. Build the memo from this page and the sheets behind it, using Template T01, and check every figure you quote against its source sheet.

### Scene 30. Investment memo: Dashboard

On screen: *Thirty nine labelled metrics by year, rounded for reading, each with its unit and its source sheet. Sheet Dashboard, range A5:H50.*

Behind the summary sits the Dashboard: a table of thirty nine labelled metrics by year, rounded for reading, each with its unit and its source sheet. Single values, such as returns, peaks, timing and readiness, sit under Year 1. Three readings teach the most. First, revenue is not cash. Cash conversion, operating cash flow over EBITDA, reads not applicable in Years 1 and 2, while EBITDA is negative. It is then minus 1.32 times in Year 3, minus 0.08 times in Year 4 and 0.25 times in Year 5, because the growing receivables book absorbs the cash. Second, the operational collection rate falls from 84.5 per cent in Year 1 to 73.3 per cent in Year 5. It is labelled not a PERFORM KPI, and the PERFORM repayment line below it reads not provided, because only company results can fill it. Third, the net FX transaction effect on costs and RBF, before price pass through, is a cost that grows from about twenty five million local currency units in Year 1 to about nine hundred and twenty eight million in Year 5. The last lines repeat the readiness result: two of twenty three gates met, and a decision of STOP, on a failed test.

### Scene 31. Twelve steps, one discipline

On screen: *Load the company's data before you trust the projections*

That completes the twelve steps. Three habits matter most. Never use an output while the master check reads error. Load the company's own history before you trust any projection. And read the Downside as carefully as the Base. The user manual, the case study, the templates and the decision tools take each step further. Thank you for watching.